

Requirements Engineering

Holiday Home Rental System

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Computing with Software Development

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Table of Contents

1.	Introduction/overview	4
2.	Functional Components	6
3.	User Requirements	7
3.1.	Holiday Home Rental System will manage Accounts	7
3.2.	Holiday Home Rental System will manage Properties	7
3.3.	Holiday Home Rental System will manage Rentals.....	7
3.4.	Holiday Home Rental System will manage Availability	7
3.5.	Holiday Home Rental System will perform administrative reporting	7
4.	System Requirements	8
4.1.	System Level Use Case Diagram	8
4.2.	Module 1 Manage Accounts	9
4.2.1.	Register Account	9
4.2.2.	Update account	12
4.2.3.	Unregister account	16
4.3.	Module 2 Manage Properties	19
4.3.1.	Register Property.....	19
4.3.2.	Edit Property	23
4.3.3.	View Property Details	27
4.3.4.	Remove Property.....	31
4.4.	Module 3 Manage Rentals.....	34
4.4.1.	Search Rental	34
4.4.2.	Create Booking	37
4.4.3.	Cancel Reservation	42
4.5.	Module 4 Manage Availability	45
4.5.1.	Add Availability.....	45
4.5.2.	Remove Availability	48
4.6.	Module 5 Manage Analysis.....	51
4.6.1.	Manage Accounts	51
4.6.2.	Yearly Revenue Analysis	54
4.6.3.	Yearly Booking Analysis	57
5.	System Model.....	60
5.1.	Level-0 DFD.....	60
5.2.	Level-1 DFD.....	61

- 5.3. Level-2 DFD (Process P1: Title) 62
- 5.4. Level-2 DFD (Process P2: Title) 63
- 5.5. Level-2 DFD (Process P3: Title) 64
- 6. Data Model (UML Class Diagram)..... 67
 - 6.1. Class Diagram..... 67
 - 6.2. Relational Schema..... 68
 - 6.3. Database Schema 68
- 7. Conclusion 71

1. Introduction/overview

A holiday home rental business has decided to automate its booking system. The new Holiday Home Rental System will support renters, hosts and an administrator. All users who wish to use the system must first register for an account. An account is identified by a unique AccountID and stores the usual contact details such as name, email, phone and status. Account records for renters, hosts and the administrator are stored in the **Accounts file**. The system allows account details to be updated when required and also supports unregistering an account by changing its status so that it can no longer be used to log in.

Hosts can register one or more properties which they offer for short-term rental. Each property is assigned a unique PropertyID and has basic details such as title, address, country, maximum guests, nightly price and status. Property records are stored in the **Properties file**. Hosts can edit the details of an existing property or mark a property as removed when it should no longer be available for booking. Renters and guests can view property details but cannot change them.

Availability for each property is managed using availability slots. An availability slot is identified by an AvailabilityID and includes the related PropertyID, a start date, an end date and a status. All slots are stored in the **Availability file**. Hosts can add new availability periods for their properties or remove existing periods when dates are no longer offered. The system uses the availability data to determine which properties are free for a given date range and to avoid double-booking.

Renters make bookings when they wish to stay in a property. To create a booking, the renter specifies search criteria such as location, dates and number of guests. The system searches the Properties and Availability files to identify properties that are available for the requested dates. The renter selects a property and confirms the booking. The system calculates the total price using the nightly rate and the number of nights, assigns a unique RentalID and saves the booking details (property, renter, dates, guests, total price and status) in the **Rentals file** with a default status of "Booked". A booking can later be cancelled by the renter, in which case the booking status is updated and the related availability is freed again. This version of the system does not process customer payments; all payments are handled outside the system.

Several administrative analysis functions are to be provided. The administrator can run a Yearly Revenue Analysis which summarises total revenue and other key figures per month for a selected year, based on data in the Rentals and Properties files. A Yearly Booking Analysis shows booking patterns such as the number of bookings and occupancy levels for each month, using data from the Rentals and Availability files. These analysis processes are reporting-only: they read from the existing Accounts, Properties, Rentals and Availability files, calculate yearly statistics and present them to the administrator without creating new persistent data.

Level 1 Processes

- P1 Manage Accounts
- P2 Process Rentals
- P3 Manage Properties
- P4 Manage Availability
- P5 Perform Analysis

Level 2 Processes

- **P1 Manage Accounts**
 - P1.1 Register Account
 - P1.2 Update Account
 - P1.3 Unregister Account
- **P2 Process Rentals**
 - P2.1 Search Rental
 - P2.2 Create Booking
 - P2.3 Cancel Reservation
- **P3 Manage Properties**
 - P3.1 Register Property
 - P3.2 Edit Property
 - P3.3 View Property Details
 - P3.4 Remove Property
- **P4 Manage Availability**
 - P4.1 Add Availability
 - P4.2 Remove Availability
- **P5 Perform Analysis**
 - P5.1 Manage Account (administrative updates for analysis)
 - P5.2 Yearly Revenue Analysis
 - P5.3 Yearly Booking Analysis

Data Stores

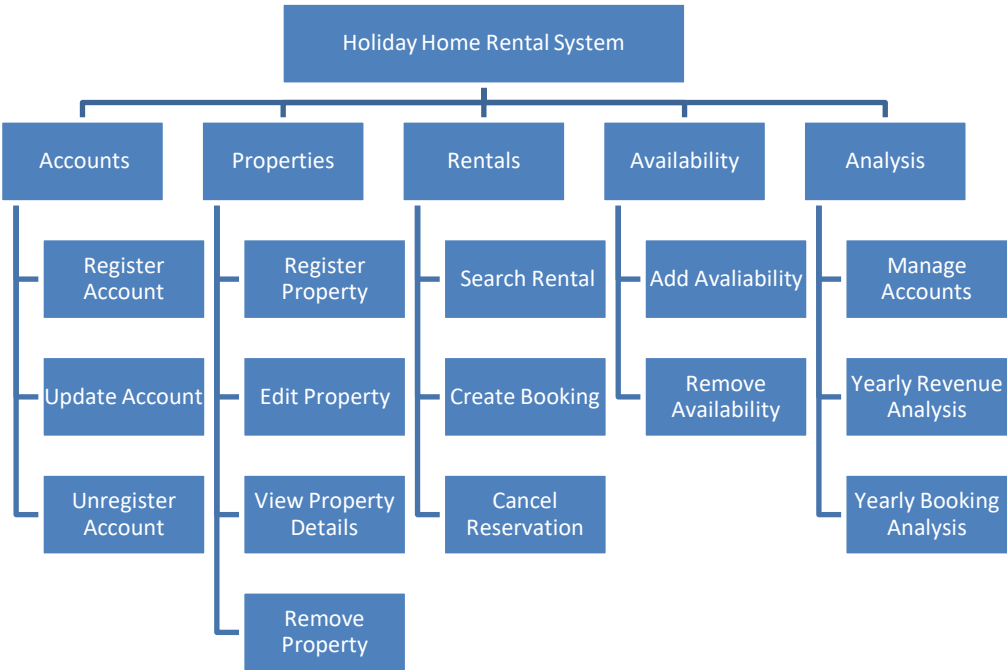
- **D1 Accounts File** – stores account records for renters, hosts and administrators.
- **D2 Properties File** – stores details of all registered properties.
- **D3 Rentals File** – stores confirmed bookings and cancellation data.
- **D4 Availability File** – stores availability slots for each property.

External Entity

- **User** – represents all external actors who interact with the system (renters, hosts and administrator).

2. Functional Components

This section presents the functional components of the proposed software system.



3. User Requirements

3.1. Holiday Home Rental System will manage Accounts

- 3.1.1. Holiday Home Rental System will add register account
- 3.1.2. Holiday Home Rental System will update account
- 3.1.3. Holiday Home Rental System will unregister account

3.2. Holiday Home Rental System will manage Properties

- 3.2.1. Holiday Home Rental System will make a register property
- 3.2.2. Holiday Home Rental System will make an edit property
- 3.2.3. Holiday Home Rental System will view property details
- 3.2.4. Holiday Home Rental System will remove property

3.3. Holiday Home Rental System will manage Rentals

- 3.3.1. Holiday Home Rental System will search rental
- 3.3.2. Holiday Home Rental System will create booking
- 3.3.3. Holiday Home Rental System will cancel reservation

3.4. Holiday Home Rental System will manage Availability

- 3.4.1. Holiday Home Rental System will add availability
- 3.4.2. Holiday Home Rental System will remove availability

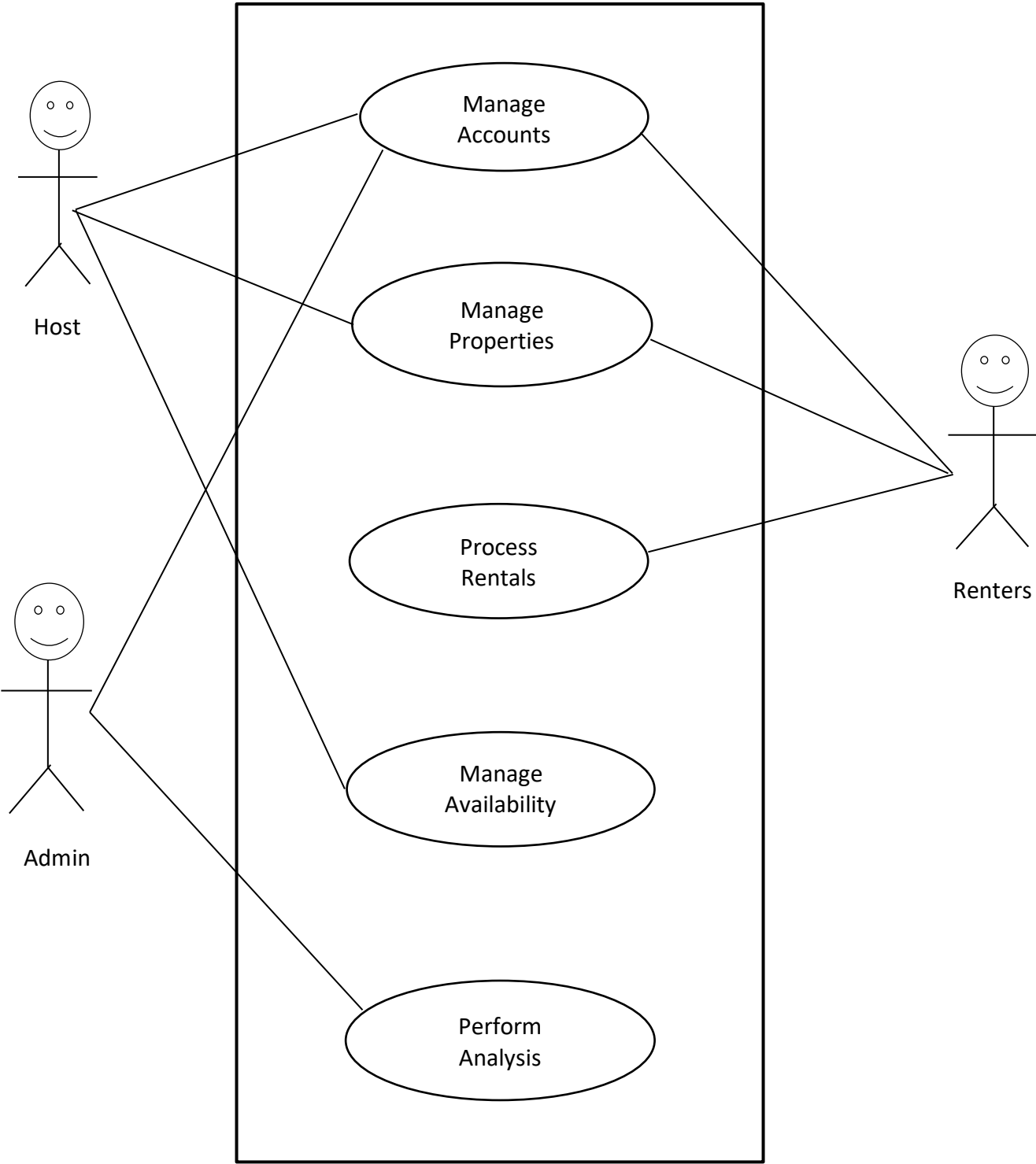
3.5. Holiday Home Rental System will perform administrative reporting

- 3.5.1. Holiday Home Rental System will manage accounts
- 3.5.2. Holiday Home Rental System will yearly revenue analysis
- 3.5.3. Holiday Home Rental System will yearly be booking analysis

4. System Requirements

4.1. System Level Use Case Diagram

The following system level use case diagram illustrates the high-level system requirements.

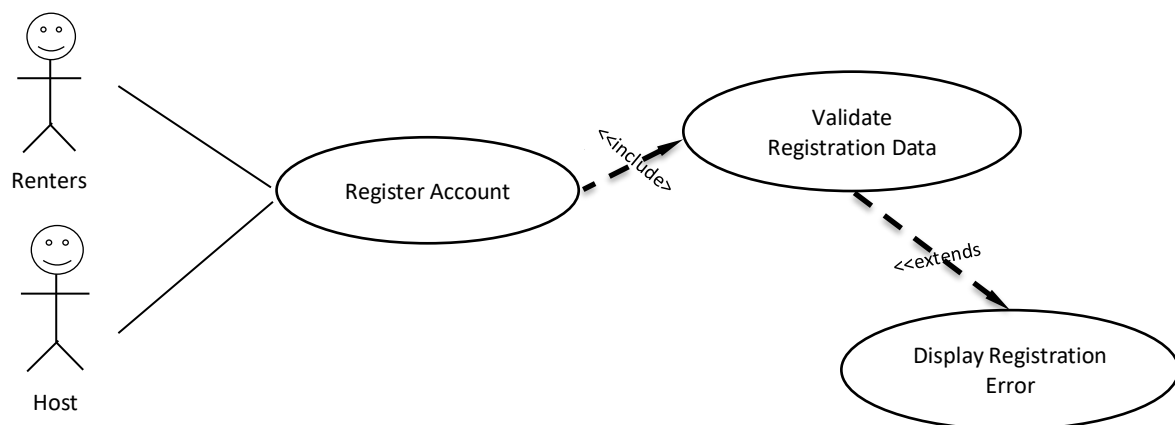


4.2. Module 1 Manage Accounts

Register account, update account, unregister account

4.2.1. Register Account

This function registers a new account in the Holiday Home Rental System. Without an account, the user cannot create or cancel reservations, and the host will not be able to manage properties and availability. On success the system stores the account with status Active and grants the chose role (Renters/Host). On behalf of users the administrator can also register accounts.



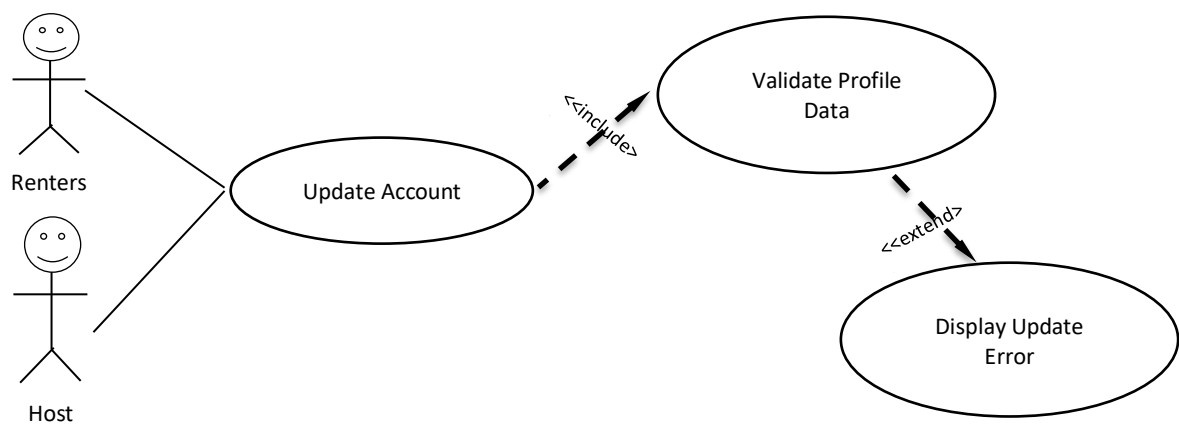
Use Case Name	Register Account	
Use Case Id	AC-01	
Priority	High	
Source	User Requirements – Manage Accounts	
Primary Business Actor	Renters/Host	
Other Participating Actors	Admin	
Description	The function creates a new user account in the Holiday Home Rental System with role Renter or Host.	
Preconditions	Renters, Host are not authenticated, Registration screen is available, System can open the Account file and Renters file.	
Trigger	User clicks Sign up/Register	
Expected Scenario	Host/Renter	System
	Step 1: Renter/Host clicks Sign up to open the registration form.	Step 2: Prepare the available Account ID and default value. Step 3: System displays the Register Account UI: First Name - varchar(30)

	Step 4: Enters all necessary data and submit. Step 5: Confirm account is to be created.	• Last Name – varchar(30) • Email – varchar(40) • Password- varchar(20) • Confirm Password • Role (Renter / Host)- char(6) • Phone – numeric(30) • Address – varchar(30) • Date of Birth – char(15) Step 6: System: • Validates the data • Email must be entered • Role must be selected (Renter or Host) • Password must match • Email must not already exists in the Accounts File. Step 7: Assign default values, for example Status = “Active”. Step 8: Account is expected to be saved in the Accounts File. Step 9: Display confirmation message: “Account has been created”. Step 10: Reset the UI.
Alternate Scenarios	Host/Renter	System

Invalid data entered		Step 6: Display appropriate error message. Step 7: Return to Step 4. Step 8: Duplicate email at Step 4 Display shows «Email already exists» or “Password do not match”.
Conclusions	A new active account is created and the user can start using the Holiday Home Rental System (optionally an initial session is started).	
Post conditions	A new active account record exists in the Accounts File with unique Email and assigned Role.	
Business Rules	• Email address must be unique in the Accounts File. • Each account has exactly one role: Renter, Host, or Administrator. • Password must satisfy the system password policy (length, complexity, etc.).	
Implementation Constraints	Implemented as a simple C# desktop application using local files. Passwords are stored as secure hashes; email verification and external identity providers are out of scope.	

4.2.2. Update account

This function lets updates Host/Renter profile details (name, phone, password, email). They can edit only their own profile, changing sensitive attributes such as role or account status is restricted to Administrator via Manage Accounts. After Host/Renter clicks «Save», system validates the input , checks email uniqueness, and sends a verification link, records the change in the audit log and update the profile.



Use Case Name	Update account	
Use Case Id	AC-02	
Priority	High	
Source	User Requirements – Manage Accounts	
Primary Business Actor	Renters/Host	
Other Participating Actors	Admin	
Description	The function is edit user profile details to the Holiday Home Rental System(name, phone, password, email). Role and account status can only be changed by the Administrator.	
Preconditions	Renters/Host is authenticated and is editing their own account.	
Trigger	Renters/Host clicks «Save profile»/ «Update account» on the profile page.	
Expected Scenario	Host/Renters	System

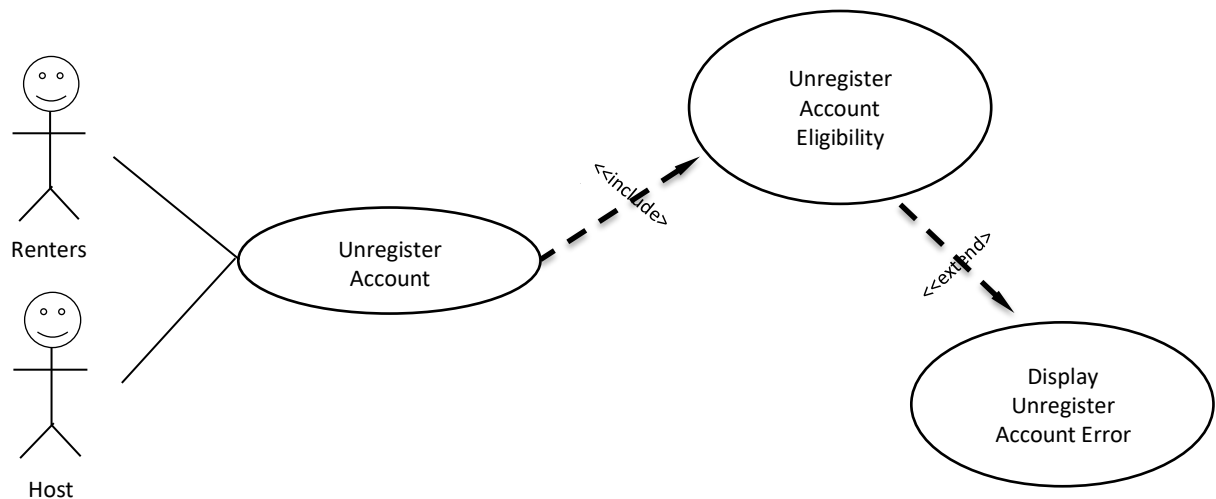
	Step 1: Host/Renters opens the profile page. Step 3: Host/Renters edits allowed fields and clicks Save.	Step 2: System reads current account data from the Accounts File (and Renters data from Renters File if used) and displays the Update account UI with the current values: • Name – varchar(60) • Phone- varchar(50) • Email- varchar(50) • Role (read-only for non-administrator) • Optional fields (for example address, date of birth) • Password change section (Current password, New password, Confirm new password) Step 4: System validates input: • Required fields are not empty • Formats are correct • Password policy is satisfied if password changed • Checks email uniqueness in Accounts File if email is changed, if email is changed Step 5: If validation is successful, system saves updated data: • Writes new values to the Accounts File for this AccountID • Writes updated renter details to Renters File if used
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		• Ignores any attempted change to Role/Status by non-administrator. Step 6: If email was changed, system marks account as “Pending email verification” and sends a verification link to the new email address. Step 7: System records an audit entry (old value, new value, time, user) and shows the message «Success – profile updated». The updated profile page is displayed.
Alternate Scenarios	Host/Renters	System
Invalid data detected		Step 4: System displays validation message(s) next to the affected fields. Step 5:: Return to Step 3. Step 6: Duplicate email detected at Step 4 Display shows «Email already exists». Step 7: Return to Step 3. Step 8: Attempt to change restricted fields (role) by non-administrator systems shows Not permitted. Step 9: Return to Step 3. Step 10: Concurrency conflict (profile changed elsewhere) at Step 5 and shows message «Profile was update by another session». Step 11: Return to Step 2. Step 12: System error to Step 5, system rolls back and shows generic error after End.

Conclusions	The account profile is updated successfully and the Host/Renter sees the new values on the profile page and can continue using the system
Post conditions	Updated profile details are stored in the Accounts File (and Renters File if used). Role and account status remain unchanged unless the change was made by an Administrator.
Business Rules	• Host/Renter can edit only their own profile. • Only an Administrator can change account role or status. • Email must remain unique in the Accounts File (if changed). • Password must satisfy the password policy when it is changed. • All profile changes must be recorded in the audit log.
Implementation Constraints	Profile updates are implemented as single file updates to the local Accounts store. Passwords remain hashed and email verification is simulated only (no real email is sent).

4.2.3. Unregister account

This function lets a Host or Renter close their own account in the Holiday Home Rental System. After the account is unregistered, the user cannot log in, create reservations or manage properties. For safety reasons the system first checks that there are no future reservations or other open obligations. Instead of removing records completely, the system changes the account status to **Inactive**, stores the date of closure and keeps history for audit.



Use Case Name	Unregister Account	
Use Case Id	AC-03	
Priority	Medium	
Source	User Requirements – Manage Accounts	
Primary Business Actor	Renters/Host	
Other Participating Actors	Admin	
Description	Unregister account a Renters/Host account. The request succeeds only when eligibility rules are satisfied.	
Preconditions	The function is to allow a Host or Renter to close their own account in the Holiday Home Rental System. The system checks for future reservations and other rules, then marks the account as Inactive	
Trigger	Renters/Host clicks in the field «Delete account».	
Expected Scenario	Host/Renters	System
	Step 1: Renters/Host opens account setting and press button «Delete account».	Step 2: System reads current account data from Accounts File and displays a UI shows a

	Step 3: Renters/Host press confirm button.	warning message and ask for confirmation . Step 4: System validates eligibility for: • No future confirmed bookings • No active properties Step 5: System change the account status to Inactive. Step 6: System: • Saves all change • Signs the user out • Shows message «Your account is Unregister»
Alternate Scenarios	Host/Renters	System
		Step 4: Future confirmed bookings are found and the system blocks the action / shows message “You have future bookings...”. Step 5: Return to Step 2 Step 6: Active property found at Step 4 and system asks to remove or archive the property Step 7: Return to Step 2
Conclusions	The account is successfully unregistered and the Renter/Host is logged out of the system.	
Post conditions	Account status is set to “Unregistered” (or “Inactive”) in the Accounts File. The user can no longer sign in or create new reservations / manage properties. History of past bookings and other audit data is retained for reporting and legal/audit purposes.	
Business Rules	• Unregister is not allowed if there are future confirmed bookings for this account. • A Host must have no active properties before unregistering. • Administrator may still access historical data even after the account is unregistered.	

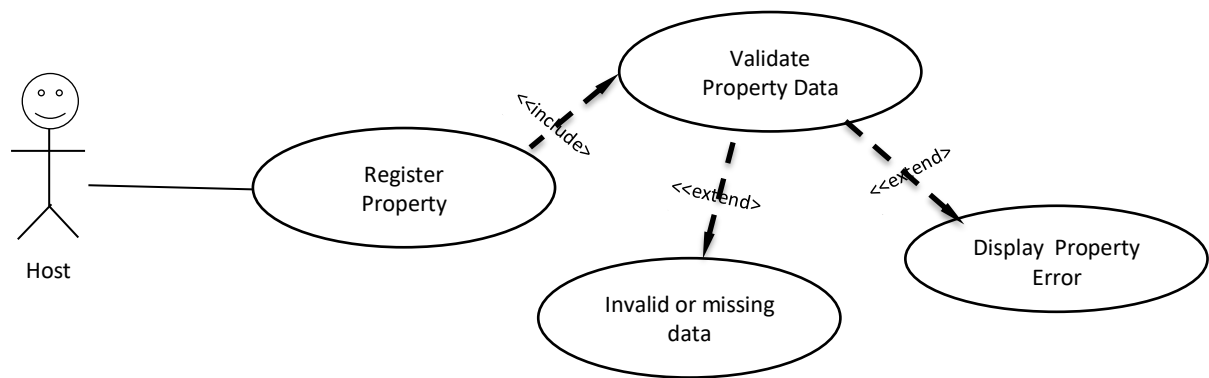
Implementation Constraints	Accounts are never physically deleted; only the Status field is changed in the Accounts file. All active sessions for this account must be invalidated after status change.
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4.3. Module 2 Manage Properties

Register property, edit property, view property details, remove property.

4.3.1. Register Property

This function registers a new property in the Holiday Home Rental System. The Host enters property details such as name, address, capacity and price. The system validates the input, assigns a new Property ID, applies default values (status and created date), saves the property in the system files and shows a confirmation message. The Host can then manage rooms, prices and availability for the new property.



Use Case Name	Register Property	
Use Case Id	PR-01	
Priority	High	
Source	User Requirements – Manage Properties	
Primary Business Actor	Host	
Other Participating Actors		
Description	The function is to create a new property record in the Holiday Home Rental System.	
Preconditions	Host is authenticated. Host is allowed to register properties.	
Trigger	Host clicks «Add property».	
Expected Scenario	Host	System

	Step 1: Host selects the Register Property function from the menu. Step 3: Host enters all required details and any optional information Step 4: Host confirms that the property should be created (clicks Save / Register property).	Step 2: System prepares the next available PropertyID and default values, for example: Status = "ACTIVE" , Created Date = today, default currency. System displays the Register Property form with empty input fields: • Property name – varchar(60) • Address city- varchar(40) • County- char(20) • Property Type-char(15) • Description- varchar(100) • Capacity-int(5) • Price – decimal(5,2) Step 5: System validates the property data: • required fields are entered • formats are correct (price is numeric and > 0, capacity is a positive integer) • checks simple business rules (for example capacity and price limits) • checks that the same host does not already have an identical property. Step 6: System checks in Properties File for a simple duplicate (same Host, same address and name).
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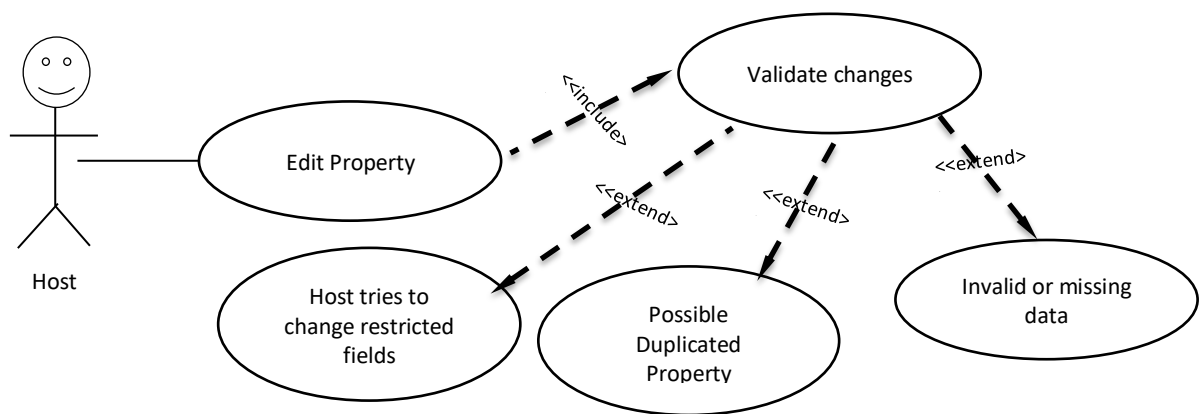
		Step 7: If validation is successful, system creates a new property record in memory with: • PropertyID – int(3) • HostID- int(3) • Property Name-varchar(50) • Address, City, Country-varchar(30) • Property Type- char(5) • Description-varchar(100) • Capacity-int(4) • Base Price per night-int(5) • Status (for example “ACTIVE”/“PendingApproval”) • CreatedDate- Date • optional fields (amenities, photos, rules). Step 8: System records this action in the system log and displays a confirmation message such as “Property ‘X’ has been registered successfully.” The property details page or host property list is displayed
Alternate Scenarios	Host	System
Invalid or missing data		Step 5: At Step 5 system finds invalid or missing values (for example empty required field, negative price, invalid capacity). Step 6: System shows clear error messages next to the wrong fields and does not save the record. Step 7: Return to Step 3 so the Host can correct the data and try again.
Possible duplicate property		Step 6: At Step 6 system finds that the new address/name combination matches another property for the same Host in the Properties File.

		Step 7: System shows message “Property already exists” and does not create a new record. Step 8: Return to Step 3 so the Host can adjust address or name if needed.
Conclusions	New property is registered successfully and shown in the Host’s property list, or the Host receives a clear message why it was not created.	
Post conditions	In normal flow a new property record exists in Properties File with a unique PropertyID and Status. If used, an initial record may be stored in Availability File .	
Business Rules	• Host can register only properties that belong to their own account. • Required address and main details must be provided. • Capacity and price must stay within limits defined by the system. • Duplicate properties (same host, same address and name) should be avoided.	
Implementation Constraints	PropertyID is generated automatically by the system and cannot be edited by the Host. Property records are written as a single update to the Properties file; if saving fails, the property is not created.	

4.3.2. Edit Property

This function lets the Host update details of an existing property in the Holiday Home Rental System. The Host can change fields such as name, description, capacity, price, amenities and photos. The Host can only edit properties that belong to their own account. Some fields such as owner or special status values may be changed only by the Administrator. After the Host clicks **Save**, the system checks the data, updates the property in the files and shows a success message.

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Use Case Name	Edit Property	
Use Case Id	PR-02	
Priority	High	
Source	User Requirements – Manage Properties	
Primary Business Actor	Host	
Other Participating Actors		
Description	The function is to edit the details of an existing property in the Holiday Home Rental System.	
Preconditions	Host is authenticated and is the owner of the property.	
Trigger	Host press button «Edit property» on the property page.	
Expected Scenario	Host	System
	Step 1: Host opens the property list or property details page and chooses Edit property . Step 3: Host make changes allowed fields.	Step 2: System reads the current property data from the Property File for this Host and PropertyID.

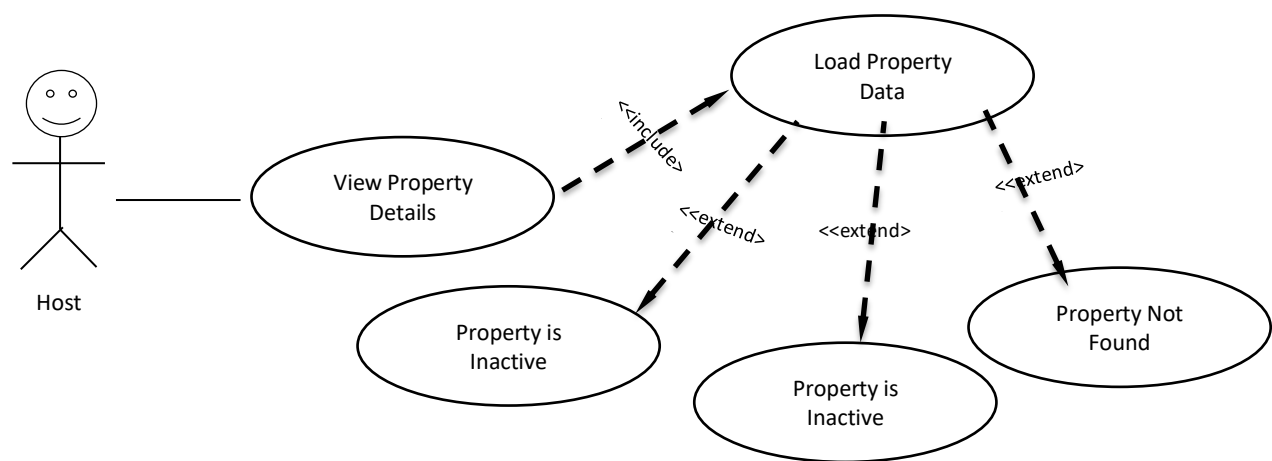
	Step 5: Host clicks Save/Update.	Step 4: System shows the Edit property screen with current values in the fields, for example: • Property Name- varchar(30) • Address, City, Country- varchar(50) • Property Type-char(20) • Description-varchar(100) • Capacity-int(3) • Base Price per night - decimal(5,2) • Status (read-only for Host)- char(6) Step 6: System checks the updated data: • required fields are not empty • formats are correct (prices and numbers are valid) • rules are followed (capacity and price within allowed limits) • if name or address changed, an optional check for duplicates for the same host is done • any changes to restricted fields (for example owner or some status values) are ignored for non-administrator. Step 7: If name or address changed, system optionally checks Properties File for a duplicate for the same Host. Step 8: System ignores changes to restricted fields (for example
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		owner or special status) for non-administrator users Step 9: If validation passes, system writes updated values to the Properties File (and to Availability File if any values affect availability). Step 10: System shows message “Property details have been updated” and displays the updated property page
Alternate Scenarios	Host	System
Invalid or missing data		Step 6: System finds invalid data (for example empty required field, price ≤ 0, invalid capacity). Step 7: System shows error messages next to the incorrect fields and does not save changes. Step 8: Return to Step 3 so the Host can correct the values and save again. Step 7: At Step 7 system finds that the new name/address combination matches another property of the same Host in the Properties File. Step 8: System shows message “Property already exists” and does not save changes. Step 9: Other valid changes can still be saved as in the main scenario.
Possible duplicate property		

Host tries to change restricted fields		Step 8: System ignores changes to restricted fields (for example owner or special status), keeps old values, and may show short message “You are not allowed to change this field.” Step 9: Other valid changes can still be saved as in the main scenario. .”
Conclusions	Property details are updated successfully, and the Host sees the new values on the property page, or no changes are saved, and the system explains what went wrong.	
Post conditions	The Property File contains the current values for this PropertyID. If used, the Availability File is still consistent with property data. The system keeps a record that this property was changed.	
Business Rules	• Only Host can edit this property. • Price more/equal than 0 and required fields must stay filled and valid. • Photos must be correct type and size.	
Implementation Constraints	Property changes are applied as a single update to the Properties file (and Availability file if needed). Large media files or advanced image processing are out of scope for this version.	

4.3.3. View Property Details

This function lets the Renter view full details of a selected property in the Holiday Home Rental System. The system reads information from the Property File and shows name, address, photos, description, capacity, price and main rules. From this screen the Renter can decide if they want to create a booking.



Use Case Name	View Property Details	
Use Case Id	PR-03	
Priority	High	
Source	User Requirements – Manage Properties	
Primary Business Actor	Host	
Other Participating Actors	Renter	
Description	The function is to display the details of an existing property in the Holiday Home Rental System.	
Preconditions	The property exists. If it is hidden/archived only show a message «Not available». Property File is available.	
Trigger	Renter clicks “View details” selects a property from list or search.	
Expected Scenario	Renters/Host	System
	Step 1: Renter selects a property from a list and clicks View details .	Step 2: System receives the selected PropertyID and reads the property data from the Property File (name, address,

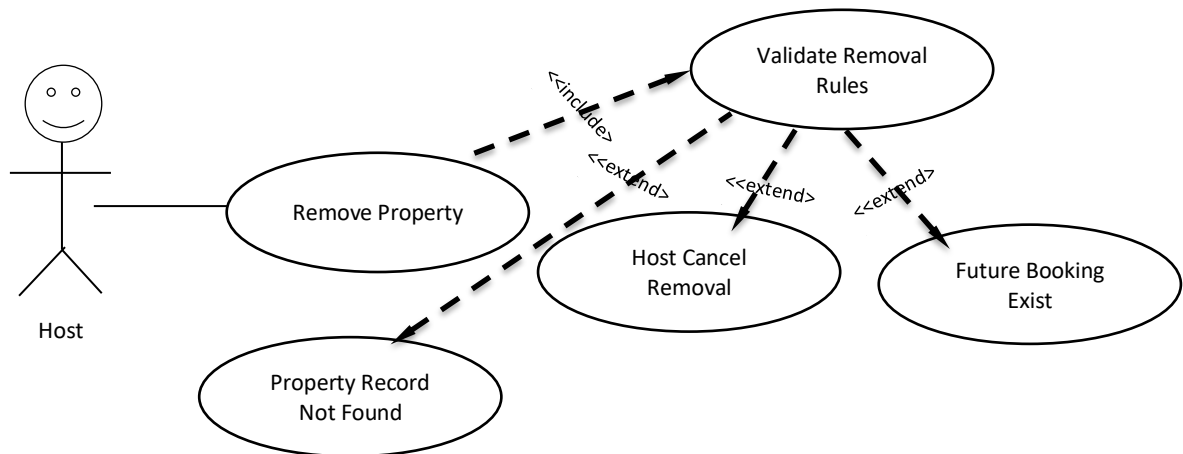
	Step 4: Renter sees the Property details screen and reads the information. Step 5: Renter enters stay dates and number of guests. Step 7: If interested, Renter clicks «Book now».	description, capacity, base price, photos, rules, status). Step 3: System displays the Property details page with current values, for example: Property Name, Address, City, Country, Property Type, Description, Capacity, Base Price per night, Amenities, Photos, House rules. Step 6: System checks the entered dates and number of guests. If data is valid, system shows an estimated price and short text about basic booking conditions. Step 8: System starts the Create Booking use case for this property.
Alternate Scenarios	Renters/Host	System
Property not found		Step 2: A system cannot find the property record in the Property File (wrong ID or removed property). Step 3: System shows a message such as “Property not available.” and returns the Renter to the previous list or search screen.
Property is inactive / hidden		Step 2: At Step 2 system finds the property but its status means it should not be shown to

Invalid dates or number of guests	Step 4: Renter enters invalid dates or selects too many guests.	normal Renters (Inactive/Hidden). Step 3: For a normal Renter, system shows message “This property is not available for booking.” and does not display full details. Step 4: If Host or Administrator opens the same property from their menu, the system allows viewing details. Step 5: System detects invalid input, shows short error message near the fields (for example “Please enter valid dates”, “Number of guests is too high”) and does not calculate price. Step 6: Return to Step 3 so the Renter can correct the input.
Conclusions	Property details are shown; the Renter can decide to create a booking. If an alternate scenario happens, no booking is started and the system explains the problem.	
Post conditions	Property data in files is not changed. Renter has viewed details for this property (only for navigation/history, no business update).	
Business Rules	Only existing properties from the Property File can be viewed.	

	• Normal Renters see only properties with allowed status (for example Active / Visible). • Inactive/hidden properties can be viewed only by their Host or by Administrator. • If dates and guests are entered, they must follow the same basic rules as for booking.
Implementation Constraints	

4.3.4. Remove Property

This function lets the Host remove a property from public use in the Holiday Home Rental System. The system checks that there are no future bookings for this property. If everything is OK, the property is marked as inactive/removed so it is no longer shown to Renters for new bookings.



Use Case Name	Remove Property	
Use Case Id	PR-04	
Priority	Medium	
Source	User Requirements – Manage Properties	
Primary Business Actor	Host	
Other Participating Actors		
Description	The function is to remove an existing property from use in the Holiday Home Rental System. For example by changing its status to Inactive/Removed.	
Preconditions	Host is authenticated. Host is the owner of the property.	
Trigger	Host presses «Remove property» / «Delete property» on the property page.	
Expected Scenario	Host	System
	Step 1: Host opens the property list or property details page. Step 3: Host clicks the Remove/Delete property button.	Step 2: Retrieve property details for this Host from the Property File and display on UI.

	Step 5: Host confirms removal (for example clicks Yes / Confirm).	Step 4: Display confirmation UI (Yes / No) showing property name and short warning text. Step 6: System checks for future bookings for this property in the Rentals File and checks relevant availability information in the Availability File . Step 7: If no future bookings exist, system updates the property status in the Property File • PropertyID – int(3) • HostID- int(3) • Status = “Removed” (or “Inactive”).-char(10) Step 8: Display confirmation message: “Property has been removed and is no longer available for booking.” Step 9: Reset UI / return to property list.
Alternate Scenarios	Host	System
Future bookings exist		Step 6: A future bookings for this property are found in the Rentals File. Step 7: System displays message “Property cannot be removed while future bookings exist.” Step 8: Return to Step 3.
Host cancels removal	Step 4: Host selects No / Cancel on the confirmation UI.	

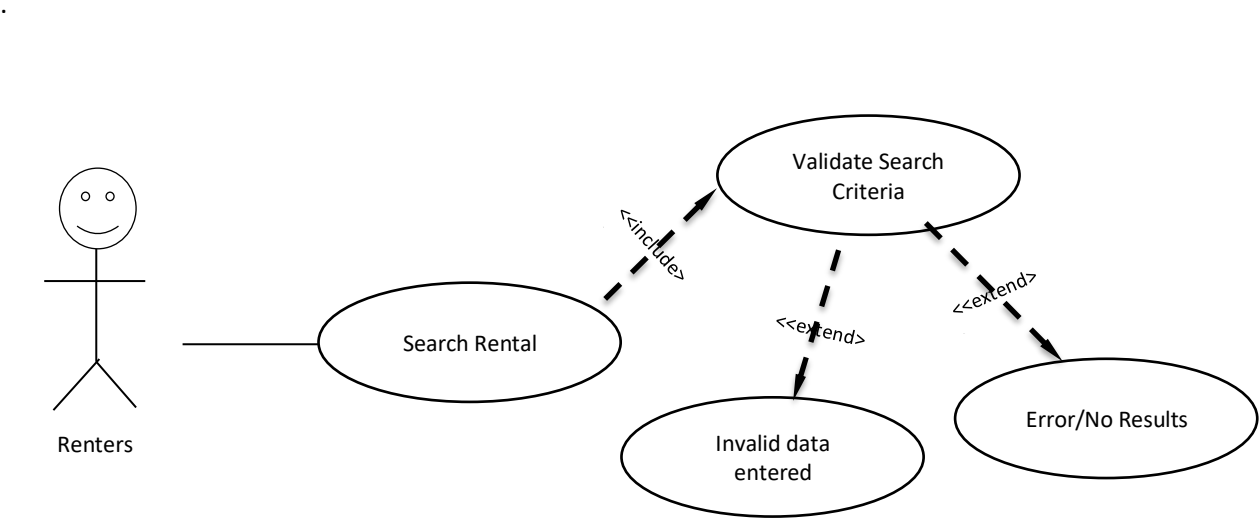
Property record not found		Step 5: System closes the confirmation UI, makes no changes and returns to the property details page. Step 2: The property record is not found in the Property File. Step 3: System displays message “Property not available – contact your administrator.” Step 4: Return to main host menu.
Conclusions	The property has been removed from use and is no longer available for new bookings, or it remains unchanged with an explanatory message to the Host.	
Post conditions	In the normal scenario the property Status in the Property File is set to “Removed/Inactive”. Property is not shown to Renters in search or booking functions.	
Business Rules	• The host can remove only their own property. • Removal is not allowed if there are future confirmed bookings. Past booking and history have to be kept for records.	
Implementation Constraints	Before changing property Status to “Removed/Inactive” the system must always check Rentals and Availability files. Any technical error during update leaves the original property record unchanged.	

4.4. Module 3 Process Rentals

Search rental, create booking, cancel reservation.

4.4.1. Search Rental

This function lets the Renter search for available rentals. The Renter enters search criteria such as location, dates and number of guests. The system validates the input, checks Properties File, Availability File and Rentals File, and returns a list of matching properties.



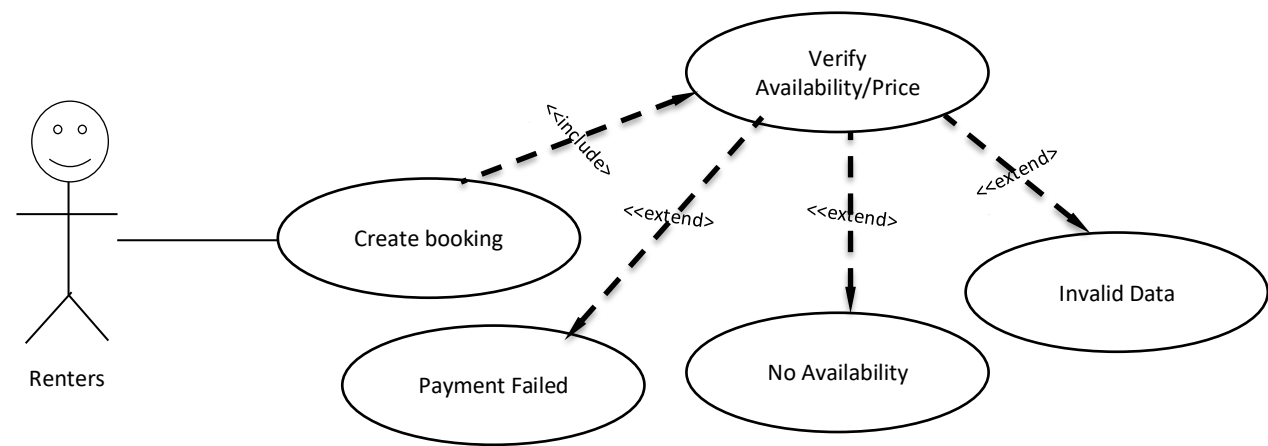
Use Case Name	Search Rental	
Use Case Id	RE-01	
Priority	High	
Source	User Requirements – Manage Rentals	
Primary Business Actor	Renters	
Other Participating Actors		
Description	The function is to search for properties that match the Renter’s criteria and are available.	
Preconditions	System is online; Properties File, Availability File and Rentals File are available.	
Trigger	Renter opens Search rentals page or clicks Search on main page	
Expected Scenario	Renter	System
	Step 1: Renter opens Search rentals page.	Step 2: System displays empty search form with fields, for example: Location, Check-in date, Check-out date, Number of guests, optional filters. Step 3: System finds properties.

	Step 4: Renter enters location, dates, guests and optional filters. Step 6: Renter clicks Search. Step 11: Renter reviews results and may click a property.	Step 5: System validates input (required fields, date order, guests > 0). Step 7: System finds candidate properties by location and status from Properties File (only Active/Visible). Step 8: For these properties system checks Availability File and Rentals File to exclude dates that are not available. Step 9: System builds and displays Search results list with matching properties (name, short description, location, estimated price). Step 10: When a property is selected, system starts View Property Details use case.
Alternate Scenarios	Renter	System

Invalid search data	Step 4: Renter enters invalid dates or leaves required fields empty.	Step 5: At system detects invalid data (for example end date before start date, missing location).
		Step 6: System shows error messages near the wrong fields and does not run search. Step 7: Return to Step 2.
No rentals found		Step 7-8: No properties satisfy the criteria and are available.
Conclusions	Renter receives a list of available rentals that match the search criteria or a clear message why no results are shown.	
Post conditions	No business data in files is changed. Optional: last search criteria may be remembered for this session.	
Business Rules	• Location and stay dates are required fields. • Dates must be a valid period (check-in < check-out). • Guests must not exceed allowed capacity when filtering properties. • Only Active / Visible properties from Properties File participate in search.	
Implementation Constraints	Search is implemented over local files only and must handle many records using simple paging/sorting. No external search engine or full-text indexing is used.	

4.4.2. Create Booking

This function lets the Renter create a new booking for a selected property. The system validates input, checks Availability File and Rentals File, calculates total price using Properties File, and then saves the booking in Rentals File. Availability is updated and a confirmation is shown.



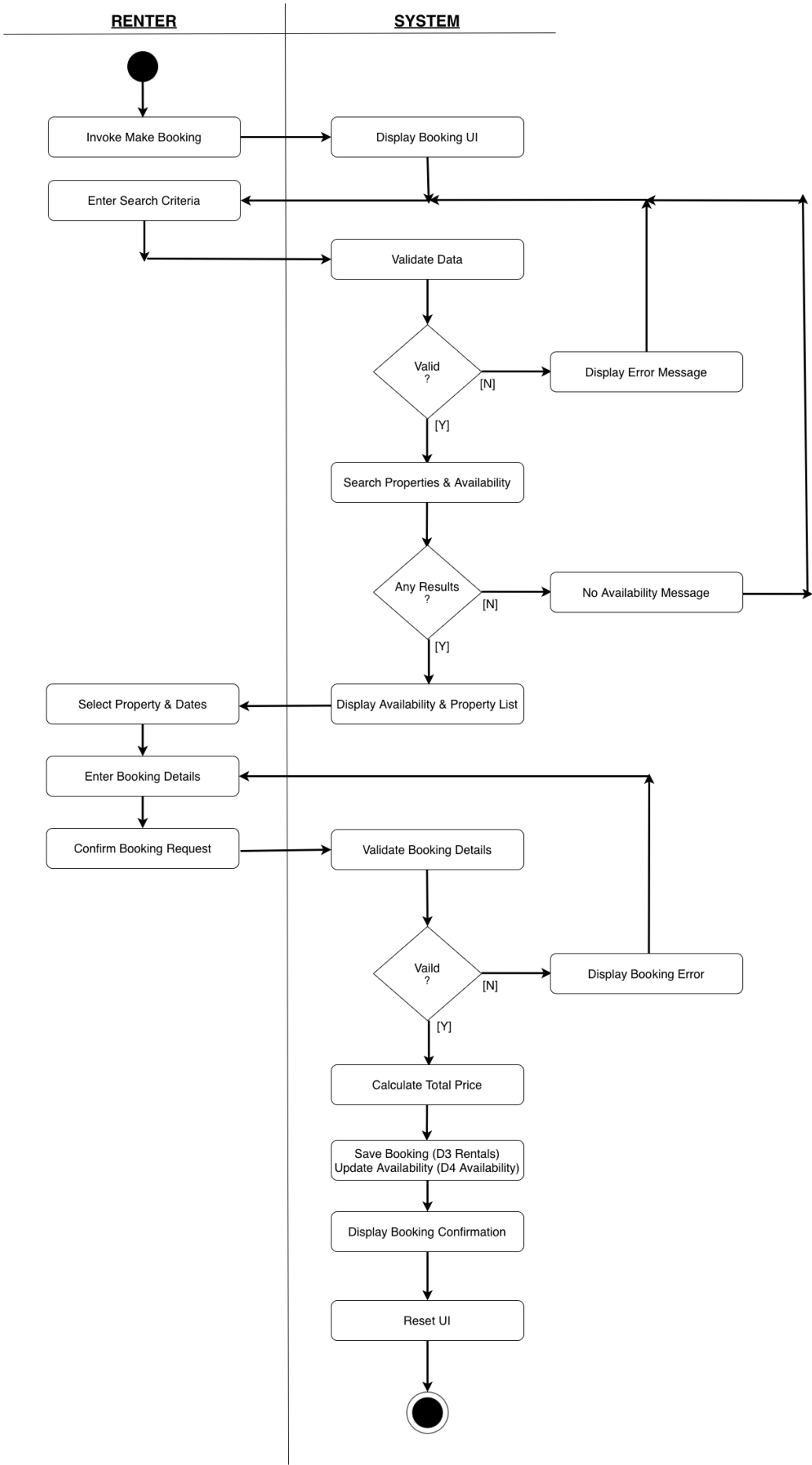
Use Case Name	Create booking	
Use Case Id	RE-02	
Priority	High	
Source	User Requirements – Manage Rentals	
Primary Business Actor	Renter	
Other Participating Actors		
Description	This function creates a new booking (rental) for a property.	
Preconditions	Renter is authenticated and account exists in Accounts File . Property exists and is allowed for booking.	
Trigger	Renter clicks “Book now / Create booking” on Property details.	
Expected Scenario	Renters	System
	Step 1: Renter starts Create Booking on a selected property.	Step 2: System reads property details from Properties File and prepares next BookingID and default values (BookingDate, initial Status). Step 3: System displays Create booking form with property summary and empty fields: Check-in, Check-out, Guests,

	Step 4: Renter enters dates, number of guests and required contact/payment data. Step 5: Renter confirms booking (clicks Confirm / Continue).	maybe contact and payment fields. Step 6: System validates data (required fields, date order, guests within property capacity, formats). Step 7: System checks Availability for the property and dates using Availability File and existing bookings in Rentals File. Step 8: If dates are available, system calculates total price using data from Properties File and, if needed, requests payment from Payment Service. Step 9: If payment is successful (or not required), system writes new booking record to Rentals File with BookingID, AccountID (from Accounts File), PropertyID, dates, guests, price, status. Step 10: System updates Availability File to mark these dates as not available. Step 11: System shows message "Booking confirmed" with booking reference and summary.
Alternate Scenarios	Renters	System

Invalid data No availability Payment failed	Step 6: Validation fails (for example dates missing, check-out before check-in, guests out of range). Step 7: System shows error messages near the wrong fields and does not continue. Step 8: Return to Step 2.
	Step 7: System finds that dates are not available in Availability File or overlap bookings in Rentals File. Step 8: System shows message “No availability for selected dates.” and does not create booking. Step 9: Return to Step 2 so Renter can change dates.
	Step 8: Payment Service returns failure or is cancelled. Step 9: System does not write booking to Rentals File and does not update Availability File. Shows message “Payment failed. Please try again or use another method.”
Conclusions	Booking is created and confirmed; dates are blocked for this property. If an alternate scenario happens, no booking is stored.
Post conditions	New booking record exists in Rentals File (normal flow). Availability File updated for booked dates.
Business Rules	Only logged-in renters (from Accounts File) may create bookings. Dates must not overlap existing bookings in Rentals File . Guests ≤ capacity from Properties File . Project rules such as minimum stay or latest booking time are applied.
Implementation Constraints	Payment is simulated; no real payment gateway is integrated in this version. Creating a booking and updating Availability must be

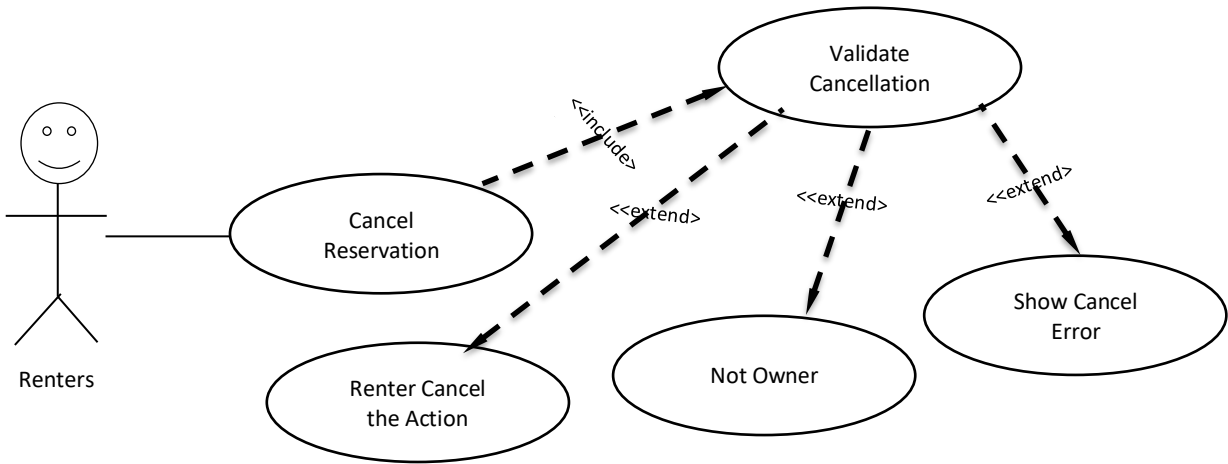
	done as one logical transaction so that either both files are updated or none.
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UML Activity Diagram



4.4.3. Cancel Reservation

This function lets the Renter cancel an existing booking. The system checks that the booking belongs to the Renter and that cancellation is allowed. If the check passes, the booking status in Rentals File is updated and Availability File is adjusted.



Use Case Name	Cancel Reservation	
Use Case Id	RE-03	
Priority	High	
Source	User Requirements – Manage Rentals	
Primary Business Actor	Renter	
Other Participating Actors		
Description	The function is to cancel a booking made by the Renter.	
Preconditions	Renter is authenticated (Account in Accounts File). Booking exists in Rentals File and belongs to this Renter.	
Trigger	Renter clicks “ Cancel reservation ” on booking details or Manage rentals.	
Expected Scenario	Renters	System
	Step 1: Renter opens the booking and chooses Cancel Reservation. Step 3: Renter confirms cancellation.	Step 2: System reads booking from Rentals File and shows confirmation UI with property, dates and current status. Step 4: System checks that booking belongs to this Renter, is active and can be cancelled according to rules (for example before check-in).

		Step 5: If allowed, system updates booking in Rentals File – sets Status = “Cancelled” and stores cancellation date/time. Step 6: System updates Availability File to free the cancelled dates (if these dates are still in the future). Step 7: System may send notification to Host and Renter via Email/Notification Service. Step 8: System shows message “Reservation has been cancelled.” and returns to Manage rentals list.
Alternate Scenarios	Renters	System
Cancellation not allowed		Step 4: System finds that cancellation is not allowed (for example stay already started, status not active, past booking). Step 5: System shows message “This reservation cannot be cancelled.” and keeps booking unchanged.
Not owner		Step 4: Renter tries to cancel a booking that is not theirs (different AccountID). Step 5: System rejects the action, shows message “You are not allowed to cancel this reservation.” and returns to Manage rentals.
Renter cancels the action	Step 3: Renter clicks No / Back on confirmation UI.	Step 4: System closes confirmation and makes no

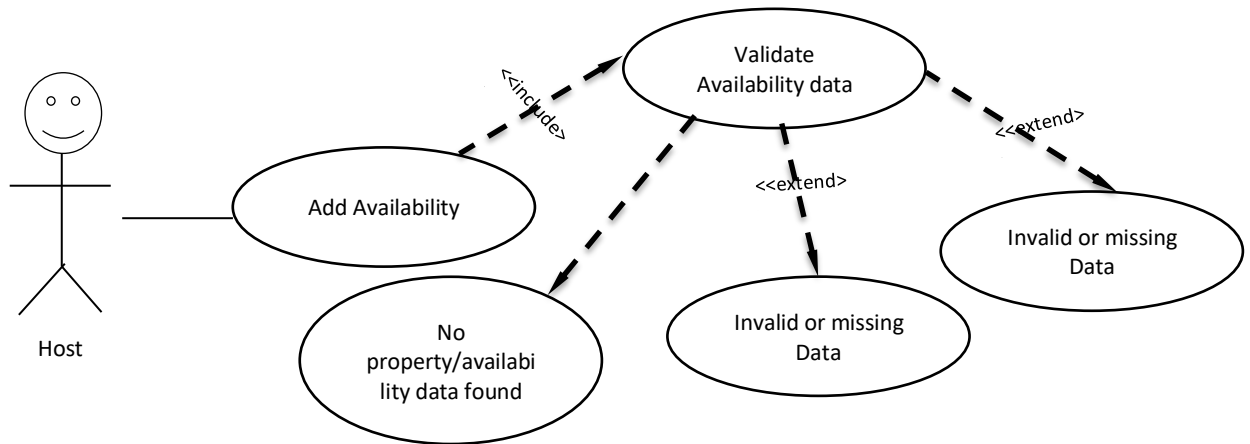
		changes; returns to booking details.
Conclusions	If allowed, booking is cancelled and future dates are released. If not allowed or an error occurs, booking remains unchanged and user gets an explanation.	
Post conditions	In normal flow: booking status in Rentals File = “Cancelled”; Availability may be updated. In alternate flows: files remain unchanged.	
Business Rules	• Only the Renter who owns the booking may cancel it (Host/Administrator use separate functions). • Cancellation is only allowed according to policy (for example X days before check-in). • Some bookings (for example already checked-in) cannot be cancelled.	
Implementation Constraints	Cancellation updates Rentals and Availability files in a single logical transaction; if any step fails, both files remain unchanged. The UI must tolerate repeated clicks on “Cancel” without creating errors.	

4.5. Module 4 Manage Availability

Add availability, Remove availability.

4.5.1. Add Availability

This function lets the Host add available dates for a property in the Holiday Home Rental System. The system validates the dates, checks existing rentals and availability, then saves the new available period in the Availability File and shows a confirmation message.



Use Case Name	Add Availability	
Use Case Id	AV-01	
Priority	High	
Source	User Requirements – Manage Availability	
Primary Business Actor	Host	
Other Participating Actors		
Description	This function adds available dates for a Host's property.	
Preconditions	Host is authenticated. Property exists in the Properties File and belongs to this Host.	
Trigger	Host presses "Add availability" on the property availability screen.	
Expected Scenario	Host	System
	Step 1: Host opens the Availability page for a property. Step 3: Host starts the Add availability function.	Step 2: Retrieve current availability periods for this property from the Availability File and display them on UI. Step 4: Display Add Availability UI with empty fields: From date, To date, optional notes.

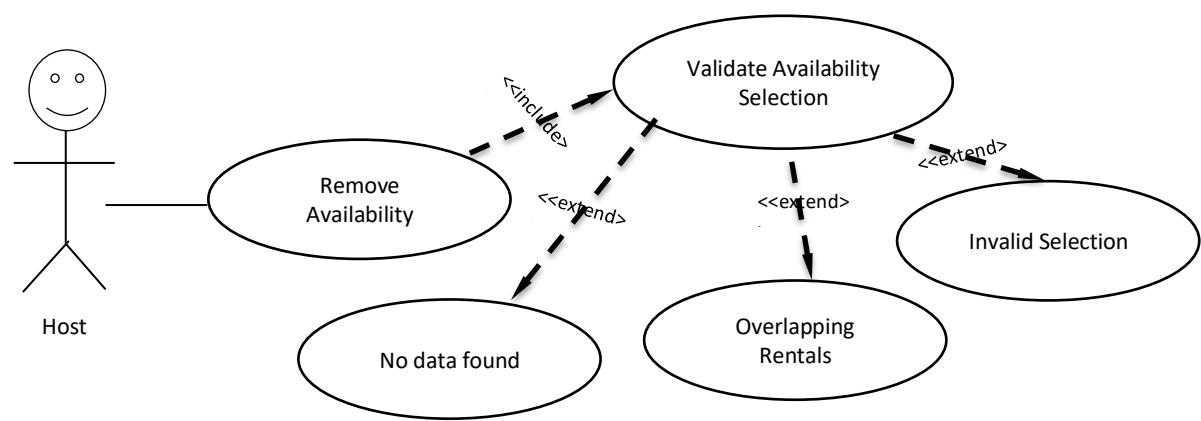
	Step 5: Host enters the required data: • From date • To date Step 7: Host confirms that the availability period is to be added.	• Step 6: Validate the data: • All fields must be entered • From date must be before To date • Period must be in an allowed range (for example not in the past). Step 8: Check Rentals File and Availability File to ensure the new period does not overlap existing rentals or existing availability blocks for this property. Step 9: Save the new availability period in the Availability File: • Property ID -int(3) • From date -Date • To date -Date • Status = "Available" .- char(15) Step 10: Display confirmation message: "Availability has been added." Step 11: Reset the Add Availability UI / return to the Availability screen.
Alternate Scenarios	Host	System
Invalid or missing data		Step 6: Validation fails (for example missing dates, From date after To date). Step 7: Display appropriate error message close to the wrong fields.

Dates conflict with existing rentals/availability No property/availability data found	.	Step 8: Return to Expected Scenario Step 4. Step 8: System finds that the new period overlaps records in Rentals File or Availability File for this property. Step 9: Display message “Selected dates are not available for adding.” Step 10: Return to Step 4 so the Host can adjust the dates. Step 2: Availability File cannot be read or property data is missing. Step 3: Display message “Availability details not found – contact your administrator.” and return to main host menu.
Conclusions	The new availability period has been saved and can be used when searching and creating bookings.	
Post conditions	A new availability record exists in the Availability File for this property and date range.	
Business Rules	• Host can add availability only for their own properties. • Availability periods must not overlap each other or existing rentals for the same property.	
Implementation Constraints	Each availability period is stored as a single record with PropertyID, From, To and Status. If the file update fails, no new period is created and a simple error message is shown.	

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4.5.2. Remove Availability

This function lets the Host remove or block existing availability for a property. The system checks that the period belongs to this property and that there are no rentals for those dates, then updates the Availability File and shows a confirmation message.



Use Case Name	Remove Availability	
Use Case Id	AV-02	
Priority	Medium	
Source	User Requirements – Manage Availability	
Primary Business Actor	Host	
Other Participating Actors		
Description	This function removes an existing availability period for a Host’s property.	
Preconditions	Host is authenticated. Property exists and has availability records in the Availability File .	
Trigger	Host presses “ Remove / Block availability ” on the property availability screen.	
Expected Scenario	Host	System
	Step 1: Host opens the Availability page for a property. Step 3: Host selects the availability period to be removed or blocked.	Step 2: Retrieve all availability periods for this property from the Availability File and display them on UI (for example as a list or calendar).

	Step 4: Host clicks Remove availability and confirms the action.	Step 5: Validate selection (a valid period is selected for this property). Step 6: Check Rentals File to ensure there are no active rentals that overlap the selected period. Step 7: If checks pass, update the Availability File for this property by removing the selected record or changing its Status to “Unavailable/Blocked”. Step 8: Display confirmation message: “Availability has been removed.” Step 9: Refresh the availability list on the UI.
Alternate Scenarios	Host	System
No period selected / invalid selection Overlapping rentals	Step 3: Host clicks Remove without selecting a valid availability period.	Step 4: System displays message “Please select an availability period to remove.” and does not update the file. Step 6: At Step 6 system finds one or more rentals in Rentals File that overlap the selected availability period. Step 7: System displays message “Availability cannot be removed because there are rentals on these dates.” and does not update the Availability File.

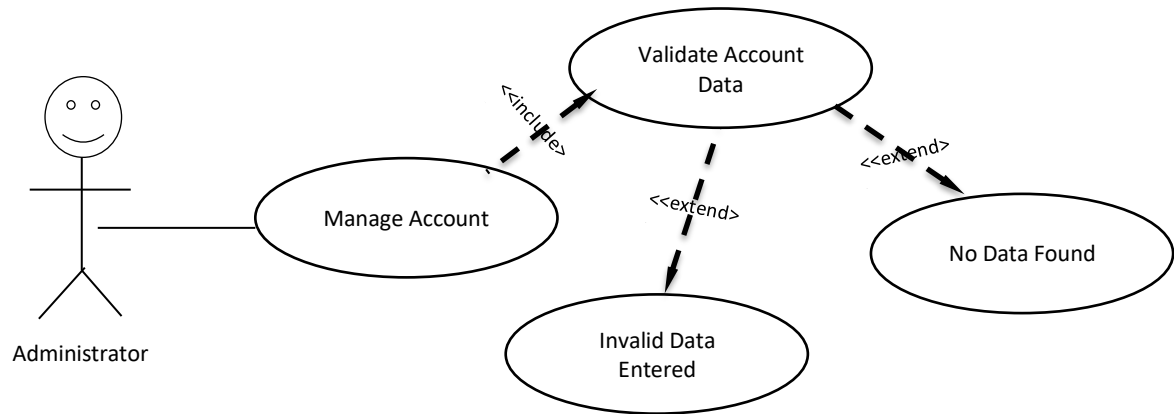
No availability data found		Step 2 or 7: System cannot read or update Availability File. Step 3 or 8: System displays message “Availability details are temporarily unavailable. Please try again later.” and returns to main host menu or previous screen.
Conclusions	The selected availability period is removed or blocked so it can no longer be used for new bookings or it remains unchanged and the Host receives a clear message.	
Post conditions	In normal flow the selected availability record is removed or its Status is set to “Unavailable/Blocked” in the Availability File .	
Business Rules	• Host may remove availability only for properties they own. • Availability that already has rentals on those dates must not be removed.	
Implementation Constraints	The system must always check Rentals file before removing or blocking availability. Updates to Availability file must be atomic; on error original records stay unchanged.	

4.6. Module 5 Perform Analysis

Manage accounts, Yearly Revenue analysis, Yearly Booking analysis.

4.6.1. Manage Accounts

This function lets the Administrator manage user accounts (Renters and Hosts) in the Holiday Home Rental System. The system reads account details from the Accounts File and allows the Administrator to view, search and update account status and role



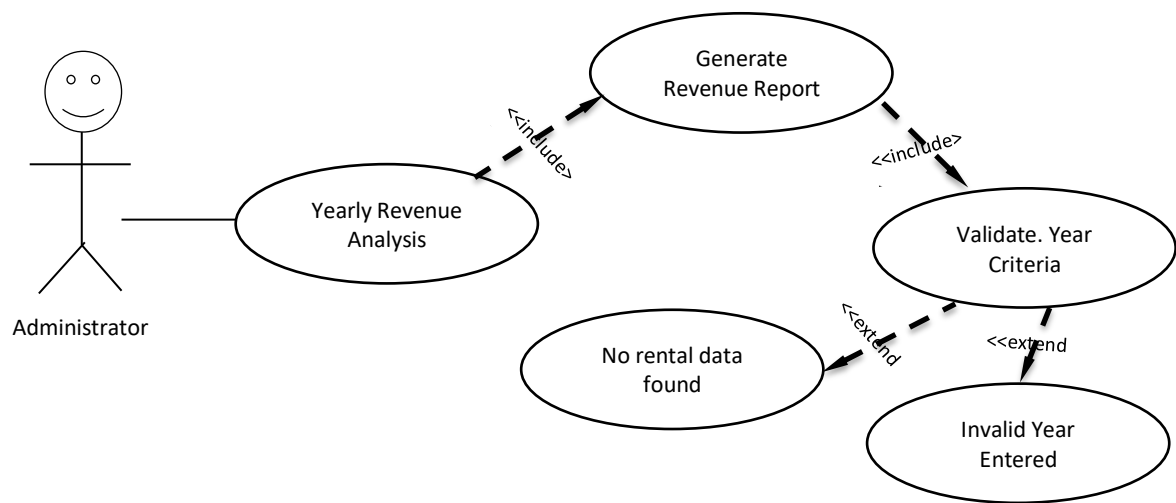
Use Case Name	Manage Accounts	
Use Case Id	PA-01	
Priority	High	
Source	User Requirements – Perform administrative reporting	
Primary Business Actor	Administrator	
Other Participating Actors		
Description	This function updates details of an existing account (status, role, contact details). AccountID cannot be changed.	
Preconditions	Administrator is authenticated.	
Trigger	Administrator starts Manage Accounts from the admin menu.	
Expected Scenario	Admin	System
	Step 1: Starts Manage Accounts function.	Step 2: Retrieve summary list of accounts from the Accounts File (AccountID, Name, Email, Role, Status).

	Step 3: Select required account (for example by clicking on a row). Step 5: Update the required data, for example: • Status (Active / Locked / Removed) • Role (Renter / Host / Admin) • Phone or email if allowed. Step 7: Confirm Save.	Step 4: Display Manage Accounts UI showing list and search/filter options. Step 6: Retrieve full details of selected account from Accounts File and load them on UI for updating. Step 8: Validate the data: • Required fields entered (email, role, status) • Email format is valid • Role is one of allowed values. Step 9: Update account details in the Accounts File: • Status – char(15) • Role-char(10) • Any other changed fields. Step 10: Display confirmation message “Account details have been updated.” Step 11: Reset UI / return to account list.
Alternate Scenarios	Admin	System

No account data found		Step 2 The Accounts File is not found or contains no data. Step 3: Display message “ No account details found – contact your administrator. ” Step 4: Return to main admin menu.
Invalid data entered	.	Step 8: Invalid data detected (for example empty role, invalid email). Step 9: Display appropriate error message near the wrong fields. Step 10: Return to Step 6.
Conclusions	The account details have been updated, or the administrator is informed why the update did not happen.	
Post conditions	Updated account details are stored in the Accounts File .	
Business Rules	Only Administrator can change account role or status. AccountID must never change.	
Implementation Constraints		

4.6.2. Yearly Revenue Analysis

This function lets the Administrator run a yearly revenue analysis. The system takes a year as input, retrieves rental records from the Rentals File, calculates total revenue and displays a summary report for that year.



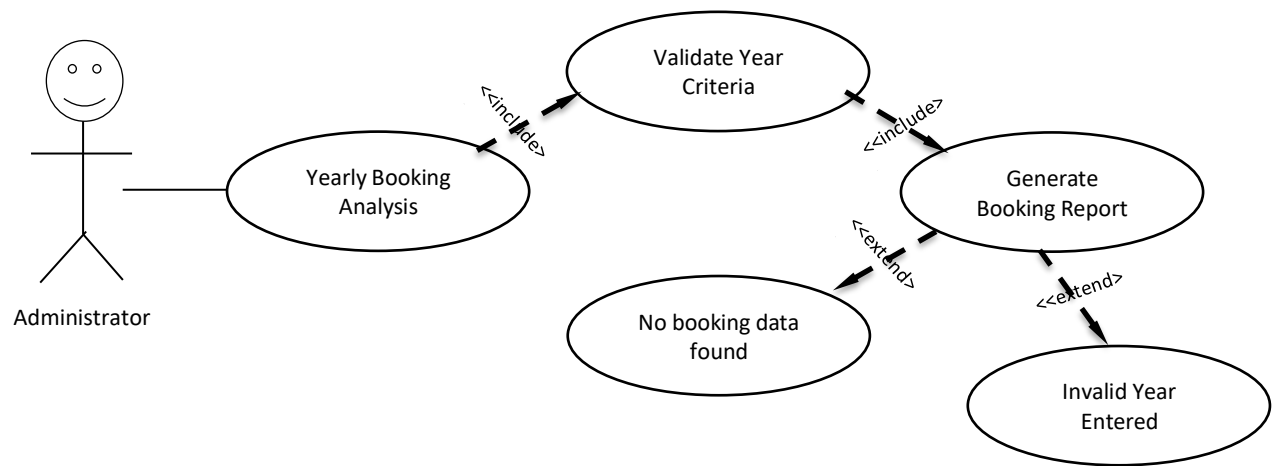
Use Case Name	Yearly Revenue Analysis	
Use Case Id	PA-02	
Priority	Medium	
Source	User Requirements – Perform administrative reporting	
Primary Business Actor	Admin	
Other Participating Actors		
Description	This function produces a yearly revenue report based on rentals stored in the Rentals File.	
Preconditions	Administrator is authenticated.	
Trigger	Administrator starts Yearly Revenue Analysis from the admin reports menu.	
Expected Scenario	Admin	System
	Step 1: Starts Yearly Revenue Analysis function. Step 3: Enter required data: • Year (YYYY)- Date • Optional filter (property or host).	Step 2: Display UI with input field for Year and optional filters (for example all properties or single property).

	Step 4: Click Run report.	Step 5: Validate the data: • Year must be four digits • Year must be in allowed range (e.g. not in the future). Step 6: Retrieve all completed/paid rentals for the selected year from the Rentals File (and property names from Properties File if needed). Step 7: Calculate total revenue for each month and/or each property according to project rules. Step 8: Display Yearly Revenue report on UI (table or chart). Step 9: Optionally allow the report to be printed or exported. Step 10: Reset UI / return to reports menu.
Alternate Scenarios	Admin	System
No rental data found		Step 6: No rentals are found in the Rentals File for the selected year. Step 7: Display message “No revenue data found for this year.” and show an empty report.
Invalid year entered		Step 5: Validation fails (for example non-numeric year or year out of range). Step 6: Display an appropriate error message and return to Step 3.

Conclusions	A yearly revenue report is produced, or the administrator is told why the report cannot be generated.	
Post conditions	No business data in the files is changed. Report output may be printed or exported.	
Business Rules	• Only rentals with suitable status (for example Completed / Paid) are included in revenue. • Currency and rounding rules for totals must be consistent across the report.	
Implementation Constraints		

4.6.3. Yearly Booking Analysis

This function lets the Administrator run a yearly booking analysis. The system uses the Rentals File (and optionally Availability File) to calculate booking counts and occupancy statistics for a selected year and displays a summary report.



Use Case Name	Yearly Booking Analysis	
Use Case Id	PA-03	
Priority	Medium	
Source	User Requirements – Perform administrative reporting	
Primary Business Actor	Admin	
Other Participating Actors		
Description	This function produces a yearly booking report (for example number of bookings per month, occupancy rates).	
Preconditions	Administrator is authenticated.	
Trigger	Administrator starts Yearly Booking Analysis from the admin reports menu.	
Expected Scenario	Admin	System
	Step 1: Starts Yearly Booking Analysis function. Step 3: Enter required data: • Year (YYYY)- char(15) • Optional filter. Step 4: Click Run report.	Step 2: Display UI with input field for Year and optional filters (property, host).

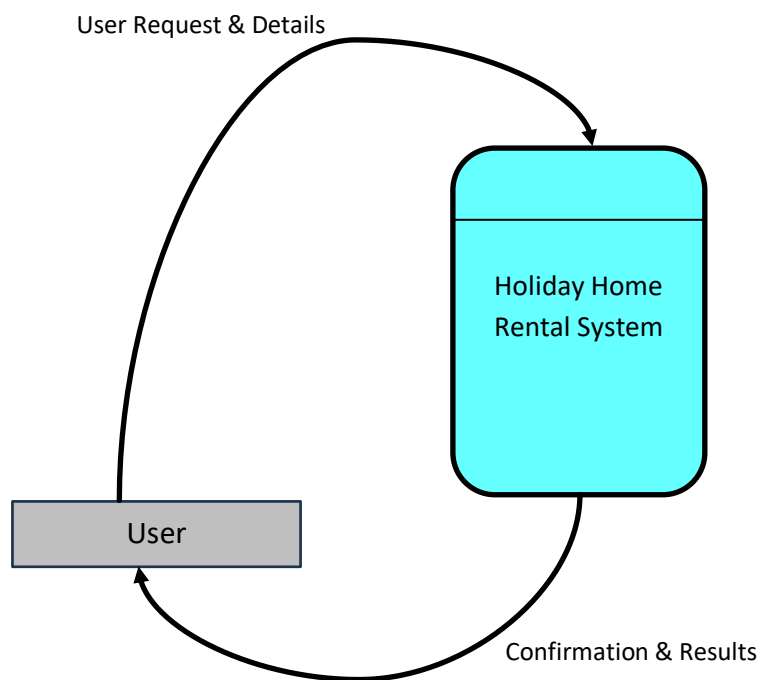
		Step 5: Validate the data (year must be numeric and in allowed range). Step 6: Retrieve bookings for the selected year from the Rentals File (and any needed data from Properties File / Availability File for occupancy calculations). Step 7: Calculate booking statistics for the year (for example number of bookings per month, nights booked, occupancy percentage according to project rules). Step 8: Display Yearly Booking report on UI (table or chart). Step 9: Optionally allow printing or exporting. Step 10: Reset UI / return to reports menu.
Alternate Scenarios	Admin	System
No booking data found		Step 6: No bookings are found in the Rentals File for the selected year. Step 7: Display message “No booking data found for this year.” and show an empty report.
Invalid year entered		Step 5: Validation fails (for example non-numeric year or year out of range). Step 6: Display an appropriate error message and return to Step 3.

Conclusions	A yearly booking report is produced, or the administrator is informed why it cannot be generated.	
Post conditions	No data in Accounts, Properties, Rentals or Availability files is changed.	
Business Rules	• Only bookings with appropriate status (for example Confirmed / Completed, not Cancelled) are counted. • Rules for calculating occupancy (e.g. nights booked / nights available) must be applied consistently.	
Implementation Constraints		

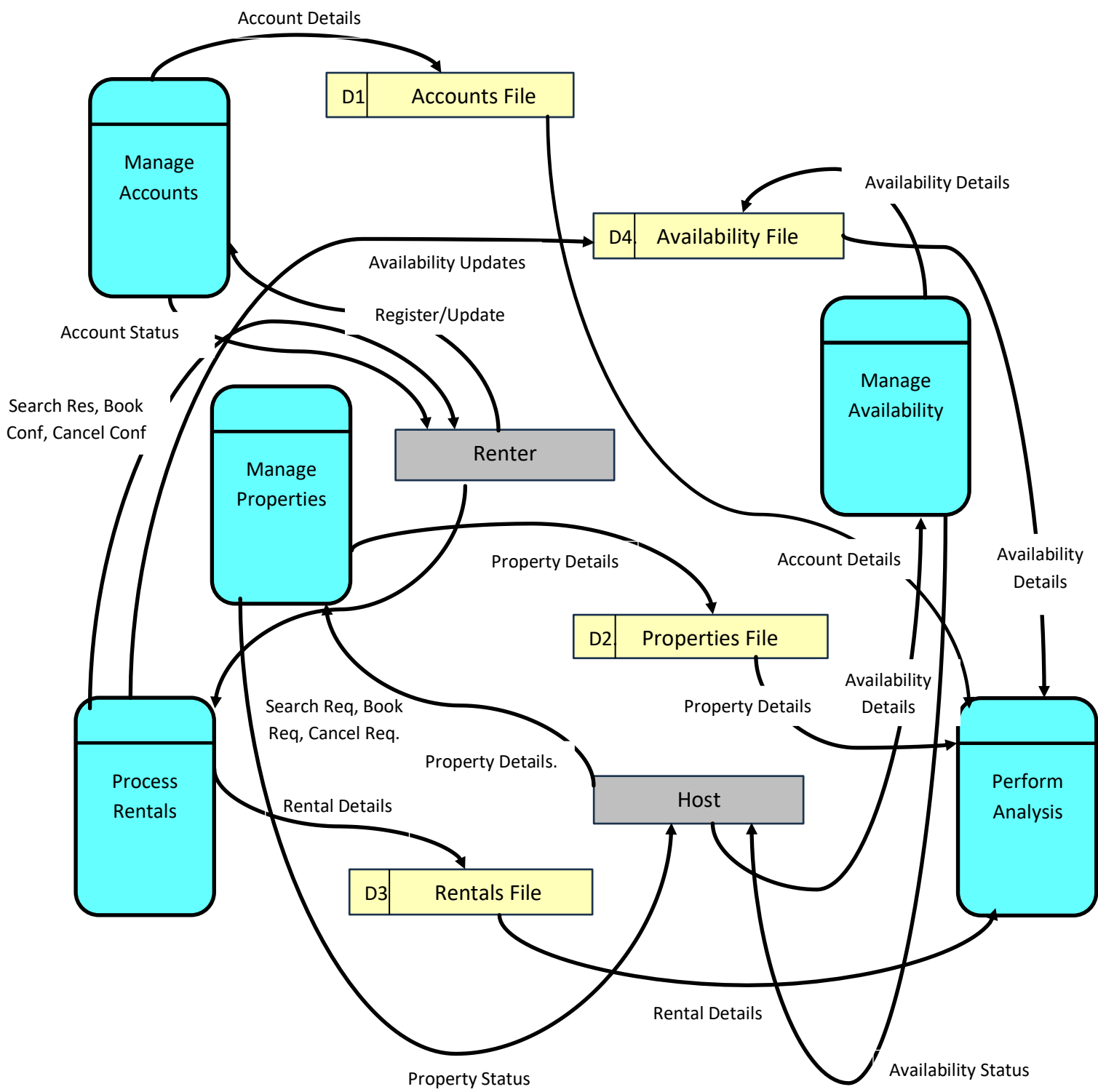
5. System Model

The following dataflow diagrams have been produced for the system:

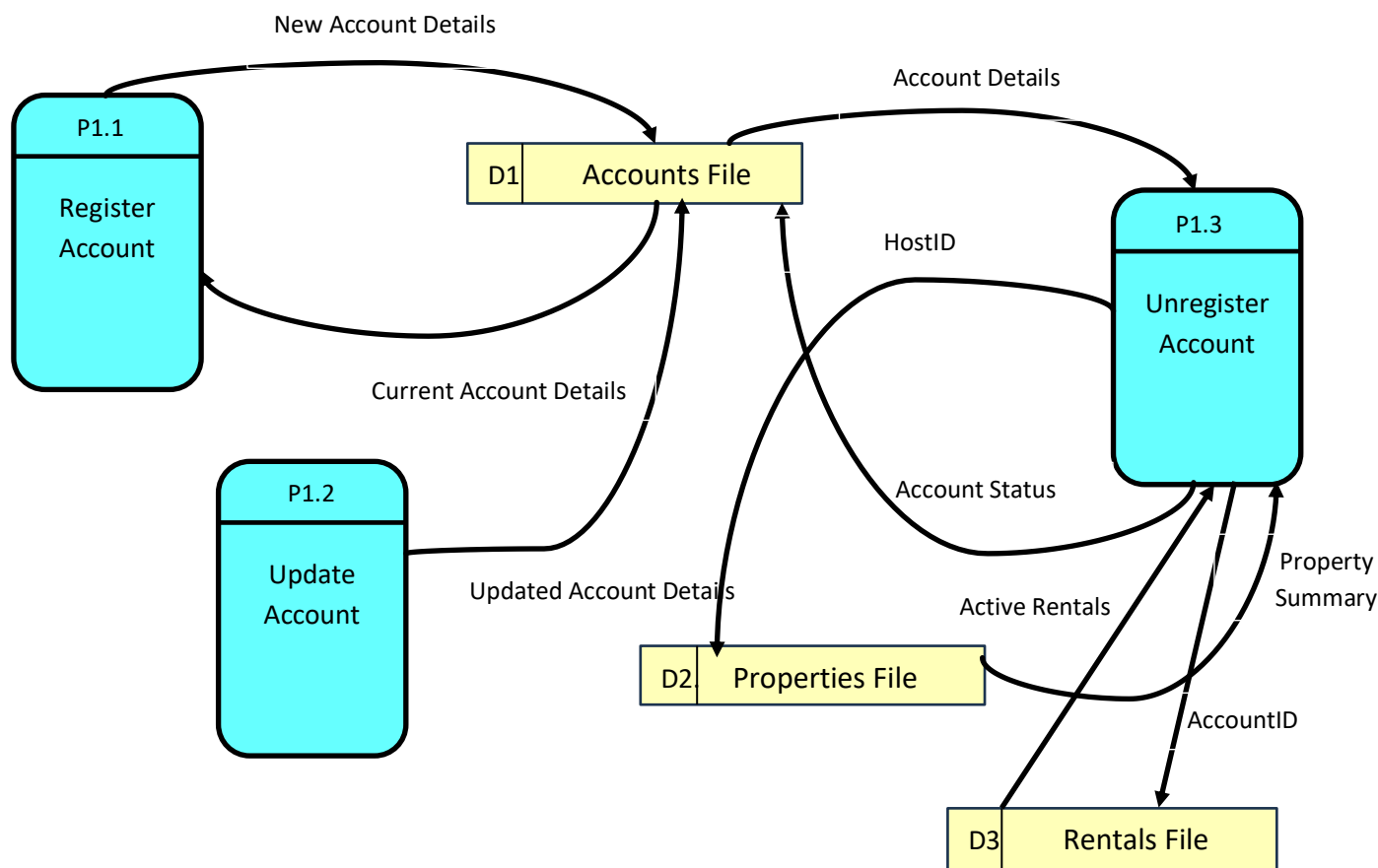
5.1. Level-0 DFD



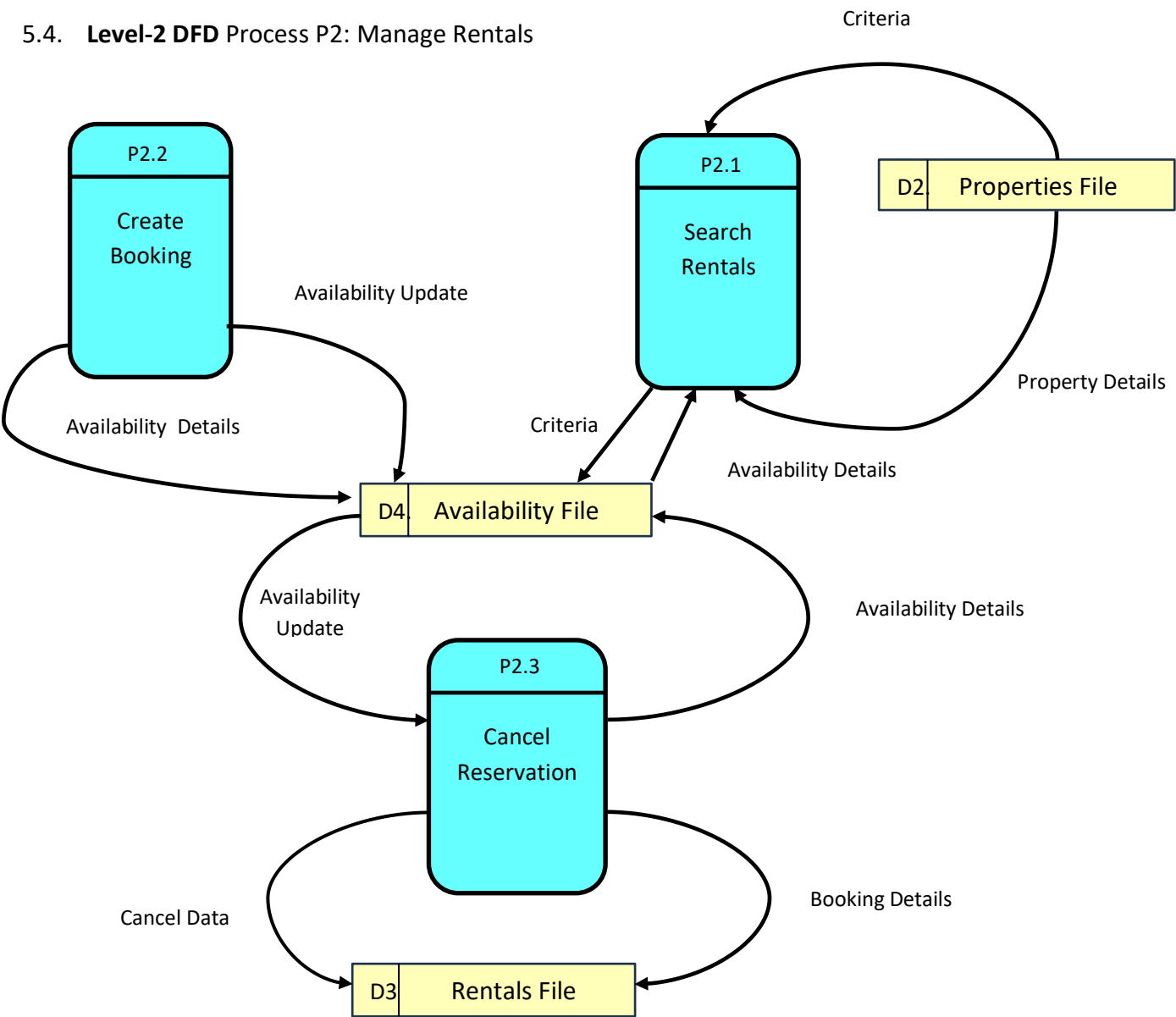
5.2. Level-1 DFD



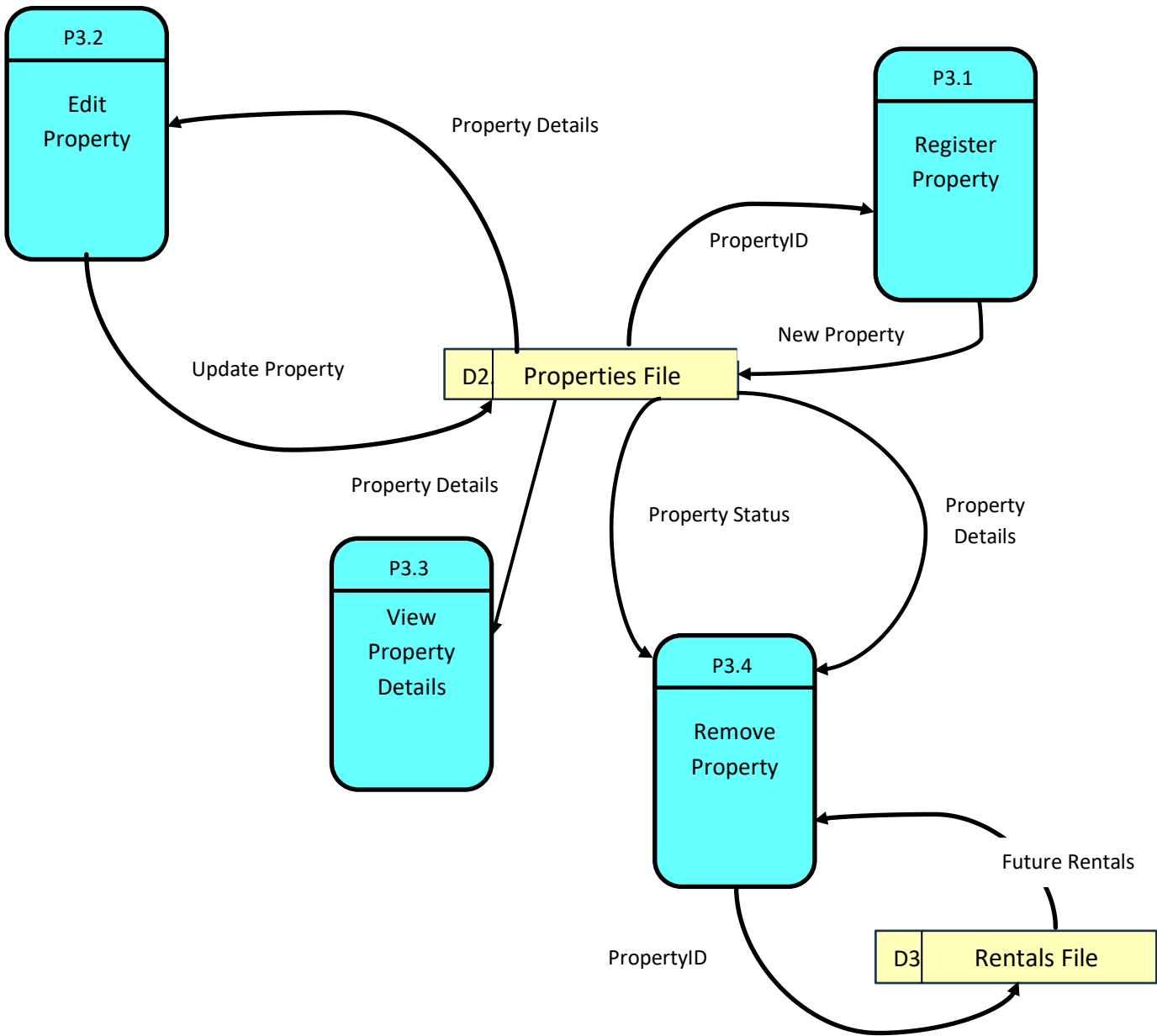
5.3. Level-2 DFD Process P1: Manage Accounts



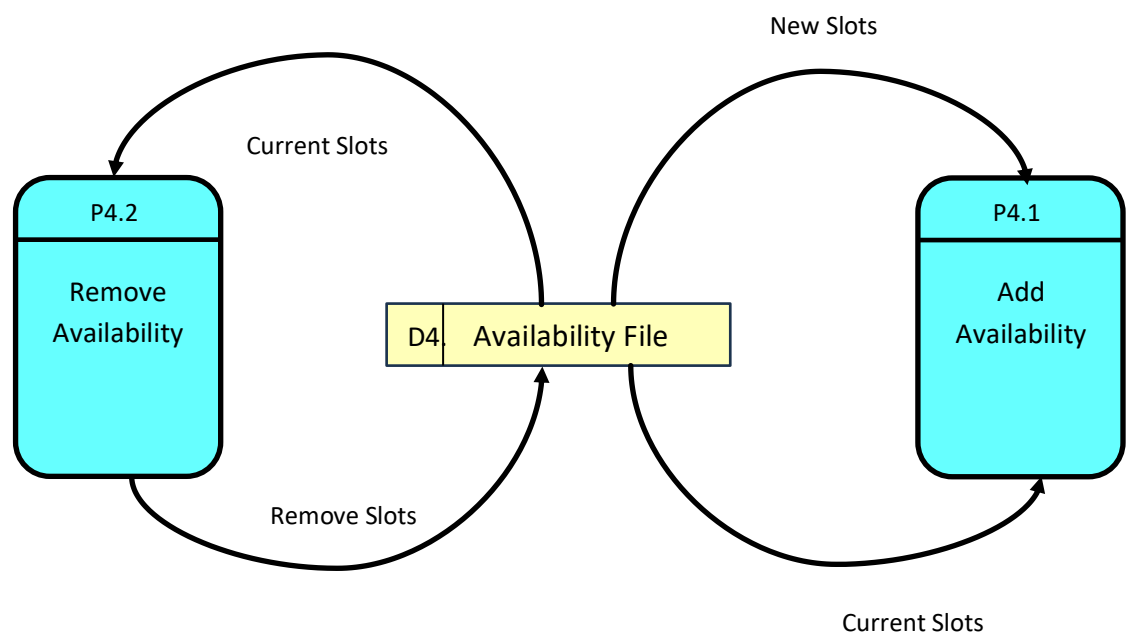
5.4. Level-2 DFD Process P2: Manage Rentals



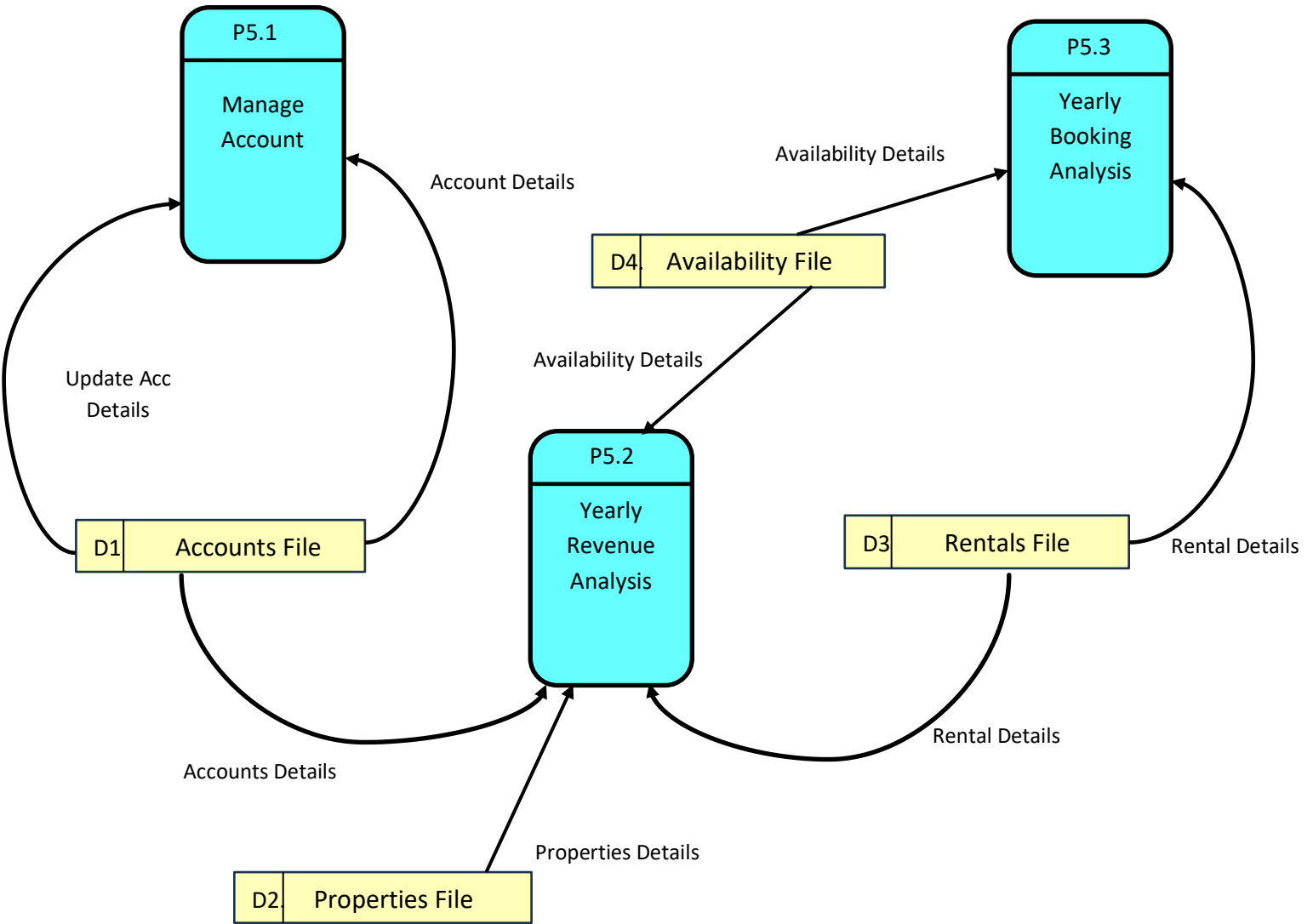
5.5. Level-2 DFD Process P3: Manage Properties



5.6. Level-2 DFD Process P4: Manage Availability



5.7. Level-2 DFD Process P5: Manage Analysis



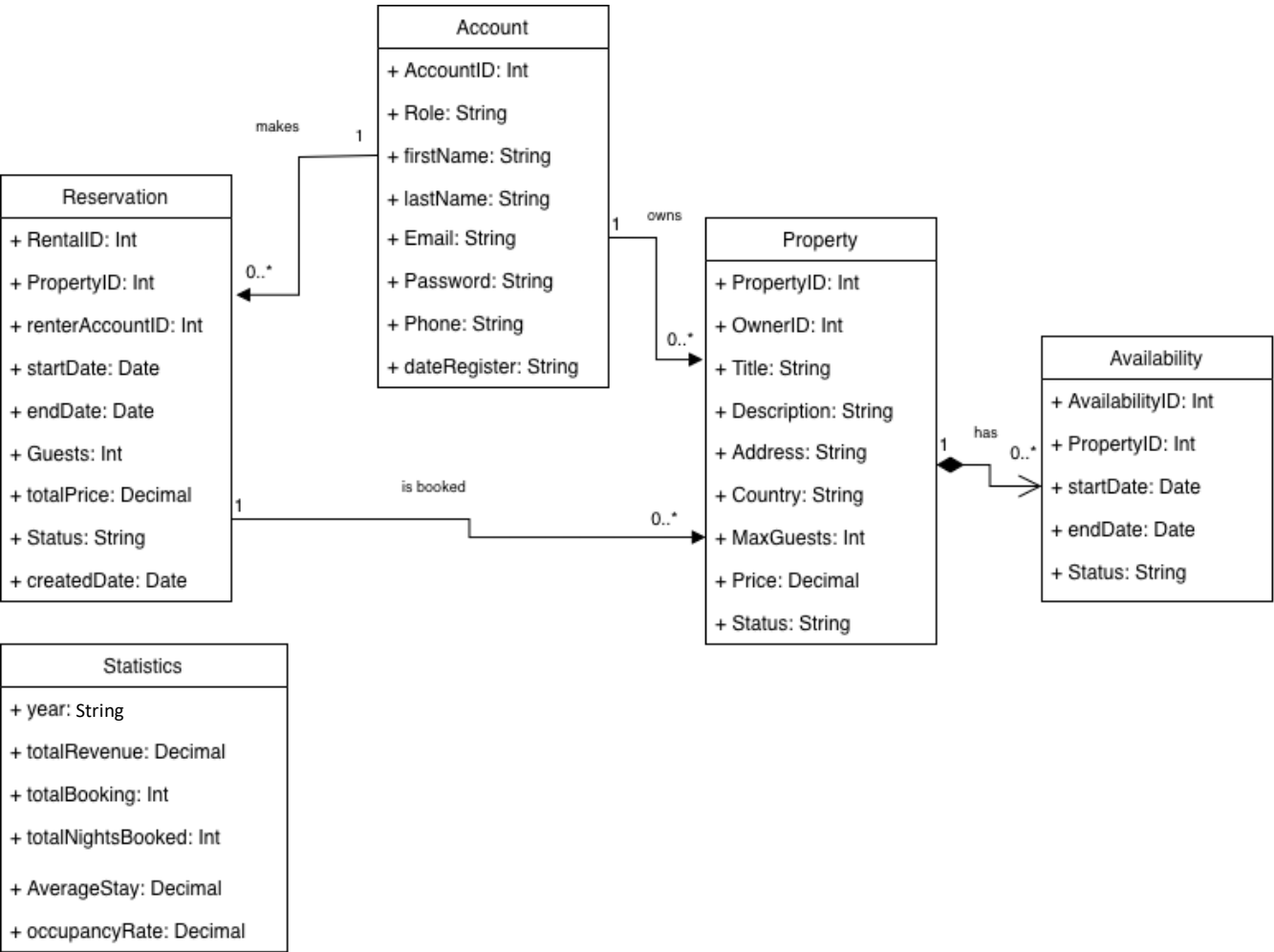
6. Data Model (UML Class Diagram)

The data model for the Holiday Home Rental System is based on five core business entities: Account, Property, Rental, Availability and Statistics.

The UML class diagram captures the static structure of the system and shows how these objects are related.

6.1. Class Diagram

Account, Property, Rental, Availability, Statistics



6.2. Relational Schema

Relational schema for the data requirements - Using *bracket notation*

Accounts (**AccountID**, Role, firstName, lastName, Email, Password, Phone, Status, dateRegister).

Properties (**PropertyID**, Title, Description, Address, Country, MaxGuests, Price, Status, OwnerID).

Rentals (**RentalID**, startDate, endDate, Guests, TotalPrice, Status, createdDate, **PropertyID**, **renterAccountID**).

Availabilities (**AvailabilityID**, startDate, endDate, Status, **PropertyID**).

Statistics (**year**, totalRevenue, totalBooking, AverageStay, totalNightsBooked, occupancyRate)

6.3. Database Schema

A definition of the database to be implemented.

This includes primary key, foreign key and other constraints to be implemented.

Schema: Holiday Home Rental System

Relation Accounts

AccountID	int(6)	
Role	char(20)	NOT NULL -- 'Renter', 'Host', 'Admin'
firstName	varchar(30)	NOT NULL
lastName	varchar(30)	NOT NULL
Email	varchar(100)	UNIQUE
Password	varchar(255)	NOT NULL
Phone	varchar(20)	
Status	char(10)	DEFAULT 'Active'
dateRegister	date	NOT NULL

Primary Key: AccountID

Relation Properties

PropertyID int(6)
OwnerID int(6) NOT NULL
Title varchar(80) NOT NULL
Description varchar(255)
Address varchar(120) NOT NULL
Country varchar(40) NOT NULL
MaxGuests int(3) CHECK(MaxGuests > 0)
Price decimal(8,2) CHECK(Price >= 0)
Status char(10) DEFAULT 'Active'

Primary Key: PropertyID

Foreign Key: OwnerAccountID References Accounts

Relation Rentals

RentalID int(6)
PropertyID int(6) NOT NULL
renterAccountID char(6) NOT NULL
startDate Date NOT NULL
endDate Date NOT NULL
Guests int(3) CHECK(Guests > 0)
totalPrice decimal(10,2) CHECK(totalPrice >= 0)
Status char(15) DEFAULT 'Booked'
createdDate Date NOT NULL

Primary Key: RentalID

Foreign Key: PropertyID References Properties

Foreign Key: RenterAccountID References Accounts

Other Constraints: CHECK(startDate < endDate)

Relation Availabilities

AvailabilityID int(6)
PropertyID int(6) NOT NULL
startDate Date NOT NULL
endDate Date NOT NULL
Status char(15) DEFAULT 'Open'

Primary Key: AvailabilityID

Foreign Key: PropertyID References Properties

Other Constraints: CHECK(startDate < endDate)

Relation Statistics

year char(15)

totalRevenue decimal(12,2) DEFAULT 0

totalBooking int(6) DEFAULT 0

totalNightsBooked int(8) DEFAULT 0

AverageStay decimal(5,2) DEFAULT 0

occupancyRate decimal(5,2) CHECK(occupancyRate >= 0)

Primary Key: Year

7. Conclusion

This system requirements document describes the proposed Holiday Home Rental System in a clear and consistent way. It starts from the user requirements and decomposes the system into five functional modules, with detailed use case narratives for each key function.

The Data Flow Diagrams (Level 0, Level 1 and Level 2) show how renters, hosts and administrators interact with the system and how the four main data stores (Accounts, Properties, Rentals and Availability) are used. The data model section then introduces a UML class diagram and derives a relational schema and database schema for the core entities: Account, Property, Rental, Availability and Statistics.

Together, these models provide a coherent description of what the system must do, what data it must store, and how the different parts of the system fit together. This specification can now serve as a solid basis for designing and implementing the database and application for the Holiday Home Rental System.